

Damage-induced basal epithelial cell migration modulates the spatial organization of redox signaling and sensory neuron regeneration

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Abstract

Summary

Epithelial damage leads to early reactive oxygen species (ROS) signaling, which regulates sensory neuron regeneration and tissue repair. How the initial type of tissue injury influences early damage signaling and regenerative growth of sensory axons remains unclear. Previously we reported that thermal injury triggers distinct early tissue responses in larval zebrafish. Here, we found that thermal but not mechanical injury impairs sensory axon regeneration and function. Real-time imaging revealed an immediate tissue response to thermal injury characterized by the rapid Arp2/3-dependent migration of keratinocytes, which was associated with tissue-scale ROS production and sustained sensory axon damage. Isotonic treatment was sufficient to limit keratinocyte movement, spatially restrict ROS production and rescue sensory neuron function. These results suggest that early keratinocyte dynamics regulate the spatial and temporal pattern of long-term signaling in the wound microenvironment during tissue repair.

eLife assessment

This **important** study identifies a novel link between the early keratinocyte response to wounds and the subsequent regenerative capacity of local sensory neurons. The evidence supporting the claims of the authors is **convincing**, although inclusion of conditional genetics or cell-autonomy tests would have strengthened the mechanistic aspects. The work will be of interest to cell and developmental biologists interested in tissue regeneration and cell interactions in a broader context.

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Introduction

Restoration of tissue function following epithelial injury requires the regeneration and activity of peripheral sensory neurons, which innervate the skin (Mullen et al., 1996 [↗](#); Rabiller et al., 2021 [↗](#); Simões et al., 2014 [↗](#)). In response to tissue damage, axonal regeneration requires the clearance of axon fragments by phagocytes followed by new axonal growth (Rasmussen et al., 2015 [↗](#)). While peripheral sensory neurons maintain a cell-intrinsic ability to regenerate following injury (Liu et al., 2011 [↗](#); Mahar and Cavalli, 2018 [↗](#); Renthall et al., 2020 [↗](#)), local environmental cues also regulate the response of sensory neurons to tissue damage *in vivo*. Recent advances have identified the contribution of supporting cell populations, paracrine biochemical signaling, and biophysical interactions in regulating peripheral sensory axon regeneration (Avraham et al., 2021 [↗](#); Cheah et al., 2017 [↗](#); Villegas et al., 2012 [↗](#)). Despite these advances, we have limited understanding of how early collective signaling in the wound microenvironment is organized and leads to sensory axon regeneration and tissue repair.

Keratinocytes are a primary constituent of epithelial tissue and play a critical role in wound healing. In addition to mediating wound closure by actively migrating toward the site of injury, keratinocytes also generate pro-reparative signals such as reactive oxygen species (ROS), which coordinate longer-term repair pathways (Dunnill et al., 2015 [↗](#); Linley et al., 2012 [↗](#); Love et al., 2013 [↗](#); Mittal et al., 2014 [↗](#); Montserrat et al., 2018 [↗](#); Sies and Jones, 2020 [↗](#)). While transient and localized ROS production promote regeneration and sensory axon regrowth, chronically elevated ROS are associated with neurodegeneration and disease (Cobb and Cole, 2015 [↗](#); Rieger and Sagasti, 2011 [↗](#)). Thus, precise temporal and spatial organization of tissue redox signaling is likely critical for efficient sensory neuron regeneration and tissue repair.

While epithelial tissue is well adapted to repair from mechanical damage, burn wounds heal poorly. Thermal injury results in chronic pain and lack of sensation in the affected tissue, suggesting that an abnormal sensory neuron response contributes to burn wound pathophysiology (Blais et al., 2013 [↗](#); Choinière et al., 1991 [↗](#); Pavoni et al., 2010 [↗](#); Tirado-Esteban et al., 2020 [↗](#); Tsolakidis et al., 2022 [↗](#)). Despite this, we lack an understanding of why sensory neuron function is impaired after burn. Our previous work has demonstrated persistent inflammation and a loss of an organized collagen matrix that impairs healing after thermal injury in larval zebrafish (LeBert et al., 2018 [↗](#), 2015 [↗](#); Miskolci et al., 2019 [↗](#)). These features recapitulate human burns and provide an *in vivo* model system to study the regeneration of sensory neurons in the wound microenvironment.

Using real time imaging, we took advantage of the optical transparency of larval zebrafish to dissect dynamic cell-cell interactions in the wound microenvironment following injury. We found that localized thermal injury induced axonal damage in the tail fin. Live imaging revealed that sensory axons are physically displaced and experience damage associated with the rapid collective migration of basal keratinocytes following burn. This early keratinocyte migration also contributes to elevated ROS at the tissue scale. Keratinocyte migration was dependent on Arp2/3 signaling and treatment with the Arp2/3 inhibitor CK666 dampened early wound-localized ROS production after burn, suggesting that dysregulated migration perturbs the temporal and spatial organization of ROS after tissue damage. Inhibiting keratinocyte migration via osmotic manipulation with isotonic solution spatially restricted ROS production and rescued sensory axon regrowth and function. Collectively, our results support the importance of regulated keratinocyte behavior for early temporal and spatial signal control that leads to sensory axon regeneration and tissue repair.

Results

Burn injury induces peripheral sensory axon damage

To visualize sensory neurons responding to tissue injury, we used 3 days post-fertilization (dpf) *Tg(Ngn1:GFP-Caax)* larval zebrafish that express GFP in sensory neurons (Andermann et al., 2002 [↗](#); Blader et al., 2003 [↗](#); McGraw et al., 2008 [↗](#)). Larvae were either mechanically injured by tailfin transection or burn as previously described (Figure 1A [↗](#)) (Miskolci et al., 2019 [↗](#)). Intravital imaging of larvae beginning at 24 hours post wound (hpw) revealed an abnormal axon morphology in burned larvae compared to mechanical transection, with axons showing fewer branch points (Figure 1B [↗](#)). To evaluate sensory axons following injury, we assessed axon density in the wounded tissue posterior to the notochord 24 hpw. Larval zebrafish caudal fins can regenerate fully by 3 days post-transection, with 60% of fin regrowth occurring by day 1.5 (Lisse et al., 2015 [↗](#)). Following transection, axon density was $89.5 \pm 0.02\%$ of the density observed in age-matched uninjured larvae 24 hpw. In contrast, we found that burned larvae had significantly reduced sensory axon density, with an axon density of $63.7 \pm 0.02\%$ compared to uninjured fins (Figure 1C [↗](#)). This relative decrease was sustained even 96 hours post wound (hpw) with an axon density of only $65.1 \pm 0.04\%$ compared to control (Figure 1C [↗](#)). To test whether this regenerative defect was associated with a defect in sensory neuron function, we assessed the touch responsiveness of wounded tissue. Light pressure was applied by an eyelash brush directly to the wound area, and sensory neuron function was scored by the presence of a tail flick reflex (Granato et al., 1996 [↗](#)). As expected, larvae wounded by transection had a nearly 100% response rate 24 hpw, indicating the rapid recovery of sensory neurons following mechanical injury (Figure 1D [↗](#)). In contrast, none of the burned larvae were sensitive to touch 24 hpw with resolution occurring by 96 hpw (Figure 1D [↗](#)). Importantly, all tested larvae exhibited a tail flick reflex when pressure was applied to the trunk, showing that the impaired sensation was limited to the damaged tissue.

We next sought to further investigate sensory neuron function in burned tissue. For this, we assessed wound-induced axonal damage using *Tg(Elavl3:GCaMP5)* zebrafish larvae that express the calcium probe GCaMP under the pan-neuronal *Elavl3* promoter. Using these larvae, we observed no noticeable change in GCaMP intensity under homeostatic conditions (Figure 1 [↗](#)–figure supplement 1A). While transient calcium increase following cell damage is required for immediate membrane repair and subsequent regeneration, chronically elevated cytosolic calcium is associated with cell degeneration and death (Khaitin, 2021 [↗](#)). GCaMP has previously been used as a marker of real-time axonal damage in zebrafish (Adalbert et al., 2012 [↗](#); Linsley et al., 2021 [↗](#); Ziv and Spira, 1993 [↗](#)). Therefore, GCaMP can be used for real-time labeling of axon damage in larval zebrafish (Adalbert et al., 2012 [↗](#); Linsley et al., 2021 [↗](#); Ziv and Spira, 1993 [↗](#)). Axon damage is characterized by the fragmenting of axons and the formation of small punctae, which are later cleared by phagocytes (Reyes et al., 2004 [↗](#); Villegas et al., 2012 [↗](#)). Accordingly, the neurotoxin sodium azide elicited widespread and long-lasting calcium-positive punctae, indicating sustained axonal damage (Figure 1 [↗](#)–figure supplement 1A) (Linsley et al., 2021 [↗](#)). Unlike *Ngn1*, the *Elavl3* promoter is expressed by both sensory and motor neurons. To ensure that the calcium increase in wounded tissue was specific to sensory axons, sensory neurons were depleted by injecting a morpholino targeting *Ngn1* (Cornell and Eisen, 2002 [↗](#)). These larvae did not have any obvious developmental defects but lacked responsiveness to touch stimulation, as previously reported (Cornell and Eisen, 2002 [↗](#)). As expected, no calcium increase was detected in *Ngn1* depleted larvae following injury (Figure 1 [↗](#)–figure supplement 1B).

Time-lapse images were taken of *Elavl3:GCaMP5* larvae to visually capture instances of axon damage indicated by calcium-positive punctae during the 30 minutes after injury (Figure 1 [↗](#)–figure supplement 1C). These images were compared to time-lapse images of sensory neuron-labeled larvae (Figure 1 [↗](#)–figure supplement 1D). In agreement with previous observations of

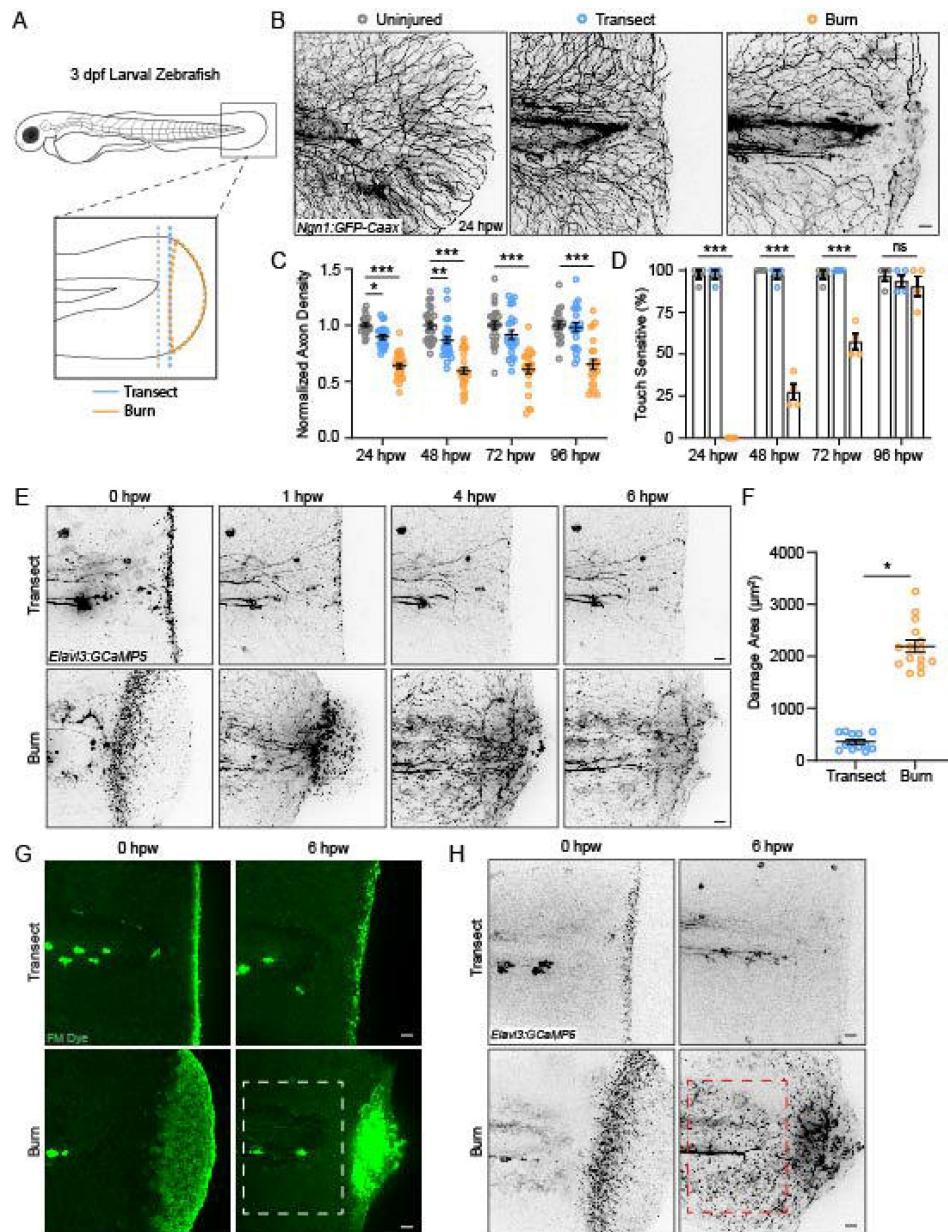


Figure 1

Peripheral sensory axons have impaired regeneration after burn injury.

(A) Schematic of larval zebrafish injury. Gray dashed line denotes area used to measure axon density to the right of the notochord. (B) Confocal max-projected images of sensory axons in uninjured, transected, and burned *Tg(Ngn1:GFP-Caax)* caudal fins 24 hpw. (C) Quantification of axon density for uninjured, transected, and burned larvae in the wound area 24-96 hpw. $N > 20$ larvae per condition from 4 replicates. (D) Quantification of sensory perception for uninjured, transected, and burned larvae 24-96 hpw. $N > 32$ larvae per condition from 4 replicates. (E) Confocal time series images of axonal damage, indicated by calcium-positive punctae (black dots), in *Tg(Elavl3:GCaMP5)* larvae following either transection or burn injury. Each series follows one representative larva over 6 hpw. (F) Quantification of axon damage area in transected and burned larvae 6 hpw. $N > 12$ larvae per condition from 2 replicates. (G) Images of larvae either transected or burned in the presence of FM 1-43 dye. White dashed box denotes area of uninjured tissue in which axonal damage appears in H. (H) Images show axonal damage following transection or burn injury. Red dashed box corresponds to the tissue region highlighted in G. In all cases, scale bars = 20 μm . * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = not significant.

axonal damage following mechanical injury (Arrázola et al., 2019; Rasmussen et al., 2015; Rieger and Sagasti, 2011), tailfin transection resulted in spatially localized axonal damage that was almost completely resolved by 1 hpw (Figure 1E). In contrast, burn injury resulted in a distinct temporal and spatial profile of sensory axon damage. While initial wound-induced sensory neuron-specific calcium increase appeared to be localized to burned tissue, axonal damage continued to increase and spread across the tissue for approximately 6 hours (Figure 1E, F). This raised the question of whether axonal damage was restricted to epithelial tissue directly impacted by injury, as observed in transected larvae. To label wounded epithelium, we used the lipophilic dye FM 1-43, which is commonly used to label damaged cell membranes after wounding (Defour et al., 2014; Lauritzen Sønder et al., 2022; McDade et al., 2021; Shannon et al., 2017). Immediately following either transection or thermal injury, axonal damage overlapped spatially with wounded epithelial tissue (Figure 1G, H). However, by 6 hours following burn injury, there was widespread damage to axons that extended beyond the initial wound area (Figure 1G, H). These findings suggest that burn injury induces axonal damage that accumulates over time and is spatially uncoupled from the surrounding epithelial damage.

The burn wound microenvironment contributes to defective axon regeneration and function

To determine if early wound signaling regulates sensory axon regeneration, we used a two-wound model (Figure 2A) to excise the burned tissue at different times post wound. In this system, zebrafish were first injured by either a primary tailfin transection or burn, and then a secondary transection injury was carried out either early at 5 minutes post-wound (mpw) or late at 6 hpw. The secondary transection was performed so that all the burned tissue was excised (Figure 2A, B). As expected, larvae that underwent only transections at both time points had full sensory function by 24 hpw regardless of the timing of the second transection injury, showing that zebrafish efficiently heal after mechanical damage (Figure 2C-E). In burned larvae, early transection after thermal injury (5 mpw) improved sensory axon regeneration and function to levels similar to transected larvae (Figure 2C-E), suggesting that burn injury does not immediately affect sensory axons differently than mechanical damage. However, when burned tissue was excised after 6 hours, significant defects were noted in both axon density and sensory function compared to larvae that either received two transection injuries or had burned tissue transected at 5 mpw (Figure 2C-E). These findings suggest that the local wound environment modulates sensory axon outcomes following burn injury, and that events in the first 6 hours after injury impact longer term sensory axon repair.

Burn injury induces the early collective movement of keratinocytes and sensory axons

To understand how burn injury damages sensory axons, we performed live imaging of epithelial keratinocytes, which closely associate with sensory axons (O'Brien et al., 2012; Rieger and Sagasti, 2011; Rosa et al., 2022). Live-imaging of *Tg(Krt4:UtrCH-GFP)* larvae that express the actin probe Utrophin under a pan-keratinocyte promoter allowed for visualization of keratinocyte dynamics following either mechanical or burn injury. In response to transection, we initially observed characteristic epithelial cell contraction at the wound edge; however, keratinocytes distal to the wound edge remained relatively stationary (Figure 3A; Supplemental Movie 1). In contrast, burn wounding resulted in a rapid collective movement of keratinocytes toward the site of tissue damage (Figure 3-figure supplement 2A). To quantify the rapid movement of keratinocytes in burn injured larvae, we used *Tg(Krtt1c19e:acGFP)* zebrafish to specifically label motile basal epithelial cells (Lee et al., 2014). Live-imaging experiments revealed that basal keratinocytes, on average, moved a total distance of $205.7 \pm 10.7 \mu\text{m}$ in the first hour following burn injury, which was significantly greater than the $58.9 \pm 5.8 \mu\text{m}$ of migration observed following

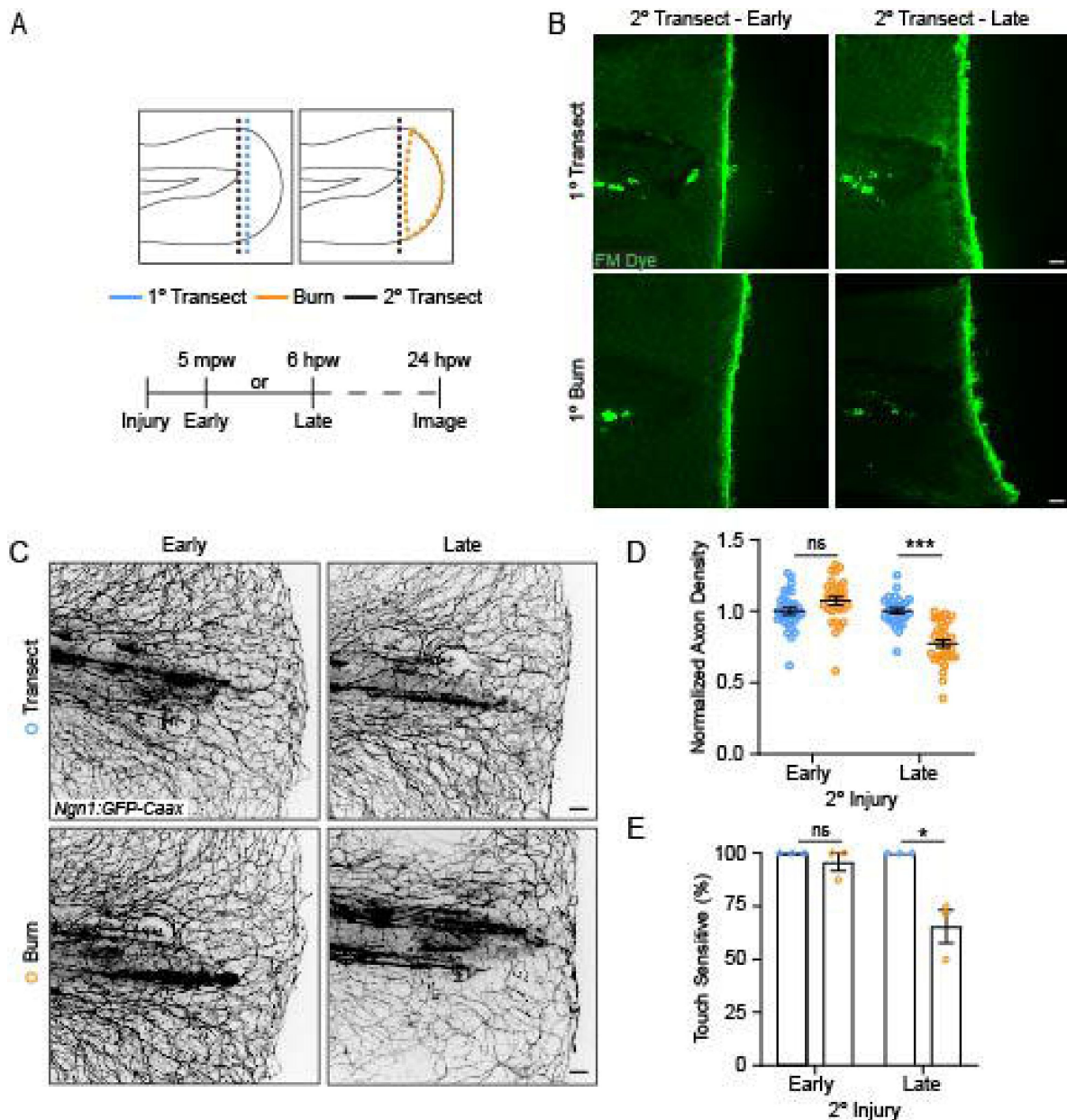


Figure 2

The burn wound microenvironment contributes to impaired sensory axon regeneration.

(A) Schematic of two-wound experiment design. (B) Confocal max-projected images of FM dye staining following secondary transection in the two-wound experiment at 5 mpw and 6 hpw. (C) Images of sensory axons in larvae subjected to an initial transection or burn injury followed by subsequent transection either early (5 mpw) or late (6 hpw). (D) Quantification of axon density in wounded tissue 24 hpw from larvae wounded as in B. $N > 28$ larvae per condition from 3 replicates. (E) Quantification of sensory perception in wounded tissue 24 hpw from larvae wounded as in B. $N = 24$ larvae each from 3 replicates. In all cases, scale bars = 20 μm . * $p < 0.05$, *** $p < 0.001$, ns = not significant.

tailfin transection (**Figure 3B** [↗](#); **Figure 3** [↗](#)–figure supplement 2A). Although keratinocytes moved as a collective in response to burn injury, they exhibited chaotic movement and appeared to be loosely associated with their neighbors.

To determine if this process was due to active migration, time-lapse imaging was performed using *Tg(Krt4:Lifact-mRuby)* larvae to visualize actin dynamics of both the superficial and basal layers. Unwounded larvae had regularly shaped keratinocytes with even actin distribution around the cell periphery in both the superficial and basal layers (**Figure 3C** [↗](#)), indicative of a non-migratory epithelium. Superficial keratinocytes in burned larvae were elongated and had an even actin distribution, but the basal keratinocytes showed actin localization to the leading edge with the formation of lamellipodia (**Figure 3C** [↗](#)), suggesting that the superficial keratinocytes are being pulled by the motile basal keratinocytes.

The axons of sensory neurons are ensheathed within actin-rich channels running through basal keratinocytes (O'Brien et al., 2012 [↗](#); Shen et al., 2019 [↗](#)). Given the chaotic and sustained keratinocyte migration associated with burn injury, we next tested if sensory axons are displaced along with associated migrating keratinocytes. Simultaneous imaging of basal keratinocytes and sensory axons following thermal injury revealed that sensory axon movement is coordinated with keratinocyte migration (**Figure 3D** [↗](#); Supplemental Movie 2). To characterize the kinetics of axonal damage following burn, time lapse movies were performed with *Tg(Elavl3:GCaMP5)* larvae to determine if the onset of axonal damage occurs during basal keratinocyte migration. Within the first hour after burn, calcium-positive punctae were identified that coincided with keratinocyte migration, indicating that early keratinocyte migration was associated with the initial axonal damage (Supplemental Movie 3). To determine if damage was limited to axons, we imaged the cell bodies of peripheral sensory neurons. At 3 days post-fertilization, the skin is innervated by Rohon-Beard (RB) and dorsal root ganglia (DRG) neurons with cell bodies that reside within and just outside the spinal cord, respectively. Following burn injury, we noted that both the RB and DRG cell bodies that innervate the caudal fin were intact and non-motile in comparison to their pre-burn morphology (**Figure 3** [↗](#)–figure supplement 1E, F). Taken together, these findings indicate that sensory axons are displaced along with the collective movement of basal keratinocytes following burn, and that this early movement is associated with the start of axonal damage.

The Arp 2/3 inhibitor CK666 impairs initial basal keratinocyte migration and modulates early ROS signaling following burn

We next determined if a known regulator of leading-edge actin dynamics and migration, Arp2/3, modulates the movement of basal keratinocytes after burn (Henson et al., 2015 [↗](#)). Treatment with the Arp2/3 inhibitor CK666 limited keratinocyte lamellipodia formation and impaired early keratinocyte migration, indicating that the early keratinocyte movement is Arp2/3 dependent (**Figure 4A–C** [↗](#)). Although early migration was impaired, by 40 mpw the migration was not significantly different between control and CK666 treated larvae, suggesting that CK666 treatment only inhibits early keratinocyte migration. To determine if this early treatment altered signaling in wounded tissue, we probed the effects of CK666 treatment on the generation of reactive oxygen species (ROS) signaling at the burn wounds. It is known that efficient tissue repair after injury relies on coordinated ROS production by epithelial cells (Yoo et al., 2012 [↗](#); Enyedi and Niethammer, 2015 [↗](#); Jelcic et al., 2019 [↗](#)). Following mechanical injuries such as tailfin transection or laser ablation, transient and localized H₂O₂ also promotes sensory axon regeneration and wound healing (Diaz et al., 2022 [↗](#); Rieger and Sagasti, 2011 [↗](#)). Because of this ROS requirement, we hypothesized that the robust keratinocyte movement and sustained damage in burned tissue may result in dysregulated ROS production. To test this, H₂O₂ level was determined using the fluorescent dye pentafluorobenzenesulfonyl fluorescein (Pfbfsf), an established readout of ROS production that has previously been used in larval zebrafish (Maeda et al., 2004 [↗](#); Niethammer et al., 2009 [↗](#)). Early after burn wounding, there was robust generation of hydrogen peroxide in burned tissue that was dampened in CK666 treated larvae. In the presence

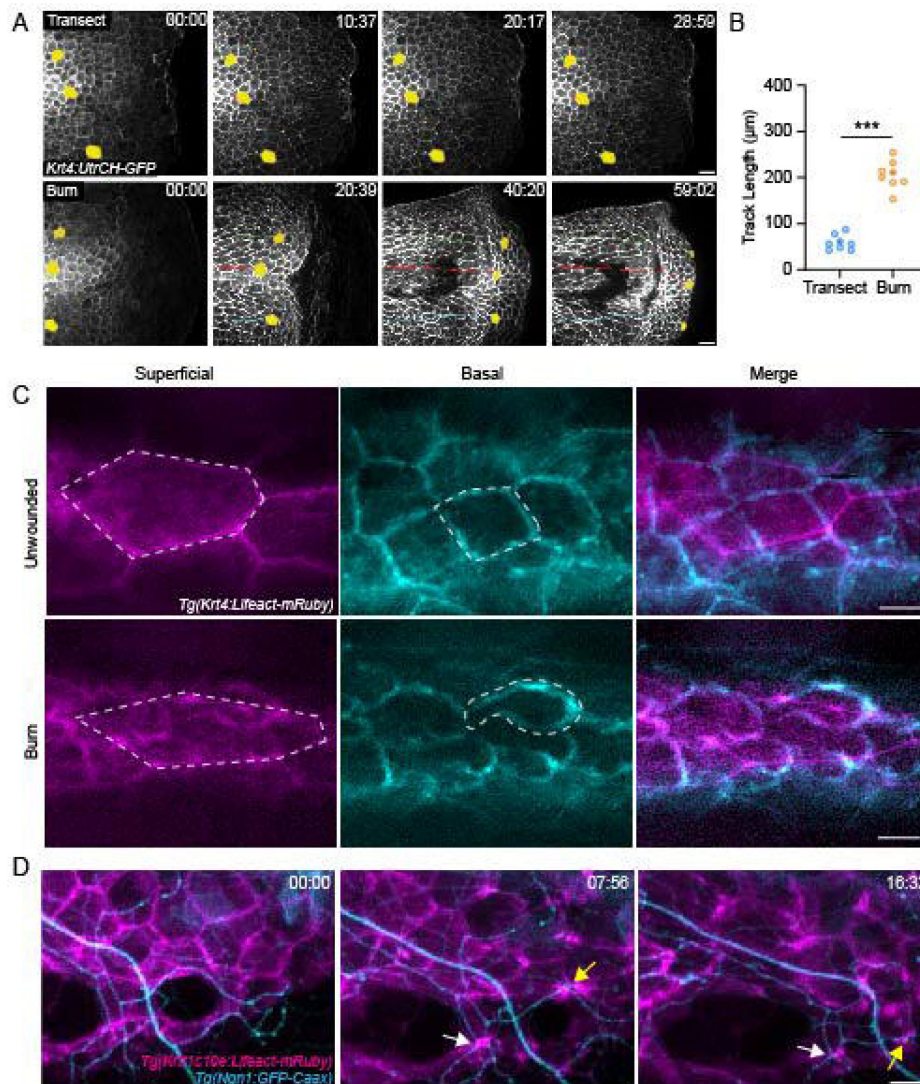


Figure 3

Burn injury induces coordinated keratinocyte and sensory axon movement.

(A) Confocal max-projected time-series images of *Tg(Krt4:UtrCH-GFP)* larvae after either transection or burn injury. Yellow pseudocolored cells and colored tracks highlight keratinocyte displacement. Scale bar=20 μ m. (B) Quantification of keratinocyte movement distance over 1 hpw. N= 8 larvae each collected from 3 replicates. (C) Confocal max-projected images of superficial and basal keratinocytes in *Tg(Krt4:Lifeact-mRuby)* labeled larvae. Left, superficial keratinocytes. Middle, basal keratinocytes. Right, Merge. Superficial and basal cell images were taken from the same z-stack and pseudocolored to match the appropriate cell layer. Dashed lines outline one individual keratinocyte. Scale bar=10 μ m. (D) Confocal max-projected time-series images of sensory axons and basal keratinocytes in dual-labeled *Tg(Krt4:Lifeact-mRuby); Tg(Ngn1:GFP-Caax)* larvae unwounded or after burn. Arrows highlight coincident movement between keratinocytes and associated sensory axons. Unless otherwise stated, scale bar=20 μ m. ***p<0.001.

of CK666, H_2O_2 production was concentrated at the wound edge, similar to what has been reported with tail transection (**Figure 4D, E**). These findings suggest that early migration alters the temporal and spatial distribution of ROS after wounding. Interestingly, CK666-treated larvae had no significant difference in axon damage or regeneration 24 hours after burn compared to controls, although there was a trend toward improved sensory function (**Figure 4F-H**). Taken together, the findings suggest that keratinocyte migration regulates early tissue scale ROS production after burn injury.

The direct effects of early ROS signaling on burn wound healing were also tested using the drug Diphenyleneiodonium (DPI) to inhibit ROS production (Li and Trush, 1998; Niethammer et al., 2009; Yoo 2012). Treatment longer than 1 hpw was not possible due to toxicity. With this short treatment, larvae showed some improvement in axon density and touch sensitivity, although it was not statistically significant (**Figure 4**—figure supplement 2B-D). These data suggest that early ROS inhibition is not sufficient to rescue axon regeneration and function, although it is possible that short term dampening of ROS may improve sensory neuron function.

Isotonic solution limits keratinocyte movement induced by burn injury and alters the temporal and spatial distribution of redox signaling

Previous studies have demonstrated that the presence of an osmotic gradient promotes keratinocyte migration via cell swelling in response to mechanical injury (Gault et al., 2014). To determine if other treatments that affect keratinocyte migration also impact tissue scale ROS production after burn, we determined if altering osmotic balance impacts keratinocyte migration and the distribution of ROS signaling after burn. Under control conditions, zebrafish are maintained in hypotonic solution. Removing this osmotic gradient by wounding larvae in the presence of solution that is isotonic to the interstitial fluid has previously been shown to inhibit keratinocyte migration following mechanical injury and impair wound healing (Gault et al., 2014; Kennard and Theriot, 2020). We found that wounding in the presence of an isotonic solution prevented the rapid movement of keratinocytes in response to a burn. Within the first hour following burn injury, keratinocyte average speed was reduced from 0.059 $\mu\text{m/s}$ in control medium to 0.003 $\mu\text{m/s}$ in isotonic medium (**Figure 5A-C**; Supplemental Movie 4). Immediately following burn injury, the level of H_2O_2 was the same between the treatment groups, indicating that cells wounded in the presence of isotonic solution maintain their normal ability to generate ROS (**Figure 5F, G**) (Eryedi and Niethammer, 2015). Examining ROS production during the first hour post-burn revealed that control larvae had increased ROS throughout the fin tissue compared to isotonic-treated larvae, while both conditions had comparable levels of ROS at the wound (**Figure 5D, E**). At 6 hours post burn, H_2O_2 production was no longer localized to the wound edge in control burned larvae and had increased throughout the tailfin. By contrast, H_2O_2 remained restricted to the wound edge in larvae burned in the presence of isotonic solution, displaying a similar localized pattern to that observed after mechanical injury (**Figure 5F**) (Jelcic et al., 2019; Korte et al., 2022). Quantification revealed that H_2O_2 level at the wound edge was similar between control and isotonic treated larvae 6 hpw. However, the level of H_2O_2 was approximately 6-fold lower in the fin epithelial tissue adjacent to the burn wound with isotonic treatment (**Figure 5F, H**). These findings suggest that robust keratinocyte movement induced by burn injury generates an oxidative environment at the tissue-scale.

Isotonic medium is sufficient to improve sensory neuron regeneration and function after burn

We next assessed sensory axon damage after burning in isotonic solution. Immediately following injury, larvae burned in isotonic solution displayed axon damage similar to larvae injured in control medium (**Figure 6A**). However, 6 hpw, axonal damage in isotonic treated larvae was

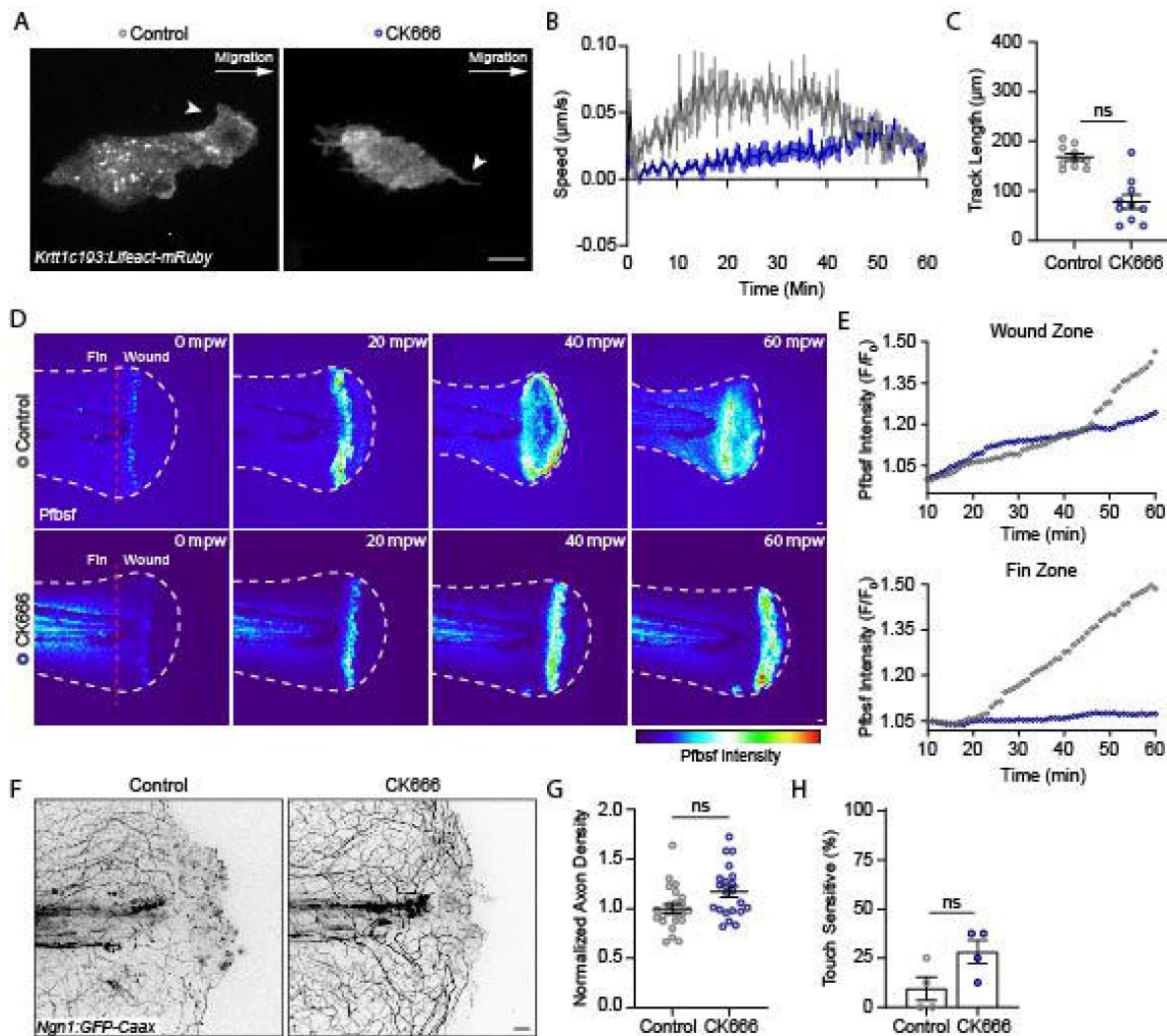


Figure 4

The Arp 2/3 inhibitor impairs early keratinocyte movement and alters the spatial distribution of reactive oxygen species signaling.

(A) Confocal max-projected images of control or CK666-treated transiently injected *Tg(Krtt1c19e:Lifeact-mRuby)* larvae. Arrows point to lamellipodia in the control larva, and lack of lamellipodia in the CK666-treated larva. Scale bar=10 μ m. (B) Plot of keratinocyte speed over 1 hpw as treated in A. N=10 larvae each collected from 3 replicates. (C) Plot of keratinocyte distance moved over 1 hpw as treated in A. N=10 larvae each collected from 3 replicates. (D) Confocal sum-projected time-series images of hydrogen peroxide level (Pfbst intensity) in 1 larva over 1 hpw in the indicated treatment. (E) Quantification of Pfbst intensity in the wound or fin area of the represented larva after burn injury as treated in D over 1 hpw. N=1 representative larva per condition. (F) Confocal max-projected images of sensory axons 24 hpw in larvae wounded in control medium or CK666. (G) Quantification of axon density 24 hpw in larvae treated as in J. N>22 larvae per condition from 4 replicates. (H) Quantification of sensory perception 24 hpw in larvae treated as in J. N=32 larvae per condition from 4 replicates. Unless otherwise specified, scale bars=20 μ m. ns=not significant.

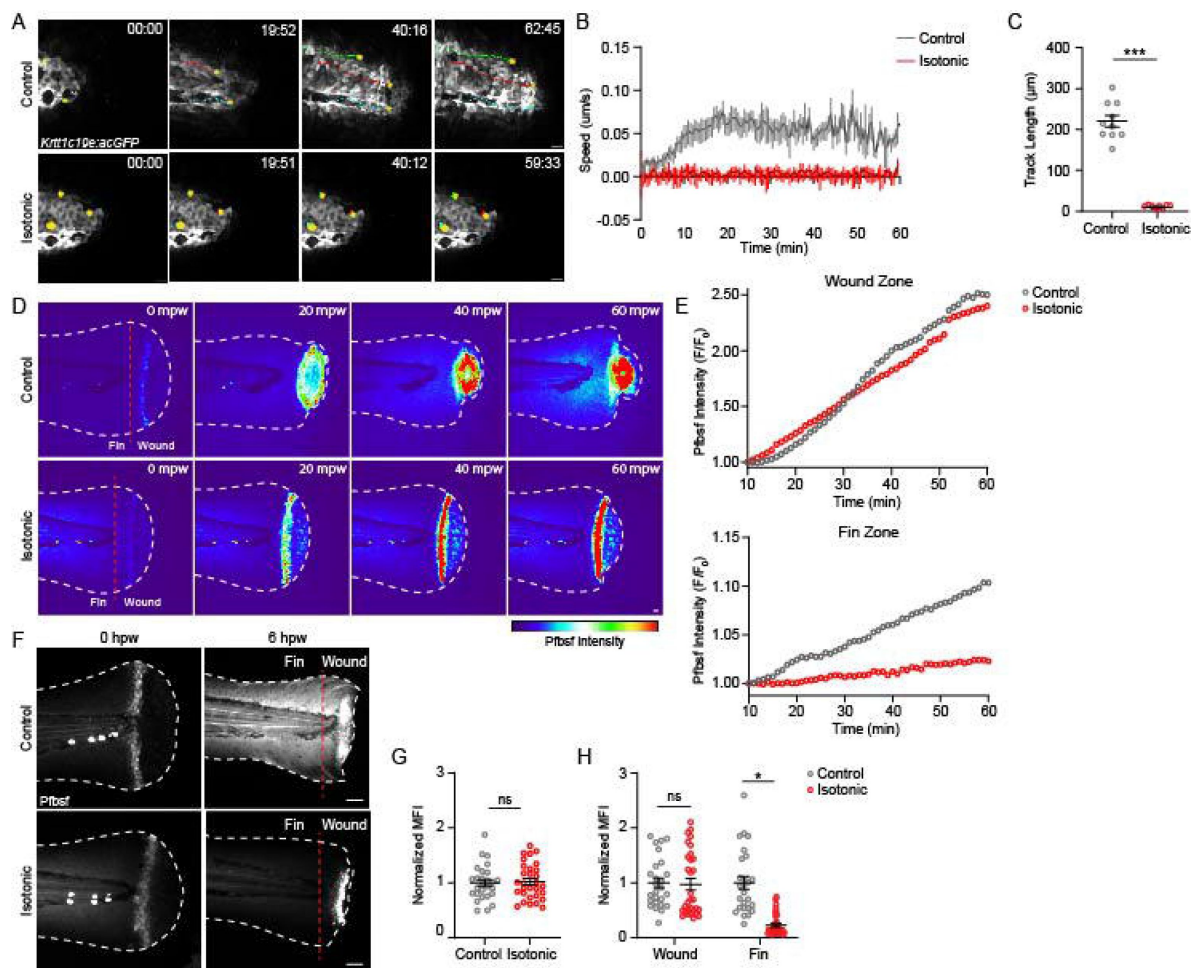


Figure 5

Treatment with isotonic solution inhibits keratinocyte migration and dampens ROS signaling.

(A) Confocal time-series images of basal keratinocyte movement in *Tg(Krtt1c19e:acGFP)* larvae over 1 hpw after burn injury in the indicated treatment. (B) Plot of basal keratinocyte average speed over 1 hpw treated as in A. N=10 larvae per condition collected from 3 replicates. (C) Distance of keratinocyte movement over 1 hpw treated as in A. N=10 larvae per condition collected from 3 replicates. (D) Confocal sum-projected, heat mapped time-series images of hydrogen peroxide level (Pfbf intensity) over 1 hpw as treated in A. (E) Quantification of Pfbf intensity in the wound or fin area of the represented larva after burn injury as treated in D over 1 hpw. N=1 representative larva per condition. (F) Confocal sum-projected images of Pfbf intensity in the fin and wound zone either 0 or 6 hours following burn injury. Dashed red line denotes the boundary between the wound area and distal fin tissue. Scale bar=50 μ m. (G) Quantification of mean Pfbf fluorescence intensity (MFI) immediately (0 hpw) after burn injury normalized to the control condition. N>27 larvae per condition from 3 replicates. (H) Quantification of mean Pfbf fluorescence intensity 6 hpw in the indicated region of the fin normalized to the control condition. N>26 larvae per condition from 3 replicates. Unless otherwise indicated, scale bars=20 μ m. *p<0.05, ***p<0.001, ns=not significant.

reduced and remained restricted to the site of injury, similar to the spatially restricted H_2O_2 signal induced by isotonic treatment (**Figure 6B**). Accordingly, larvae burned in isotonic medium had significantly greater axon density 24 hpw, and more than 85% of isotonic treated larvae had normal sensory function by 24 hpw (**Figure 6C-E**). To determine if the benefit of isotonic solution was due to its ionic composition, we tested the effects of an isotonic solution of the sugar D-Sorbitol. We also found that an isotonic solution with D-Sorbitol limited basal keratinocyte migration and had normal axon density and sensory function 24 hpw, suggesting that the benefit of isotonic solution is independent of effects on electrical cues (**Figure 6**–figure supplement 3A-C).

To determine if early keratinocyte migration contributes to the impact of isotonic solution on sensory neuron function at later time points, we treated with isotonic solution starting 1 hour following burn injury (**Figure 6F**). When isotonic medium was added 1 hpw, after keratinocyte migration was complete, there was no rescue of ROS production in either the wound area or the fin 6 hpw (**Figure 6** G, H). Additionally, there was no improvement in sensory axon density or function 24 hpw, supporting the idea that early wound events during the first hour are critical for their effects on later sensory neuron function (**Figure 6I-K**). Collectively, these findings suggest that early keratinocyte movement after burn coordinates spatial redox signaling and impacts sensory axon regeneration.

Discussion

Tissue repair requires the coordination of signaling across spatial and temporal scales. Our prior work has shown that early ROS signaling immediately after mechanical damage is necessary for longer term tissue repair (Yoo et al., 2012). Wound-induced ROS production is also required for leukocyte recruitment, ECM remodeling, and sensory axon regeneration in response to tissue injury (LeBert et al., 2018; Niethammer et al., 2009; Rieger and Sagasti, 2011; Yoo et al., 2011). While the requirement of ROS production following tissue injury is clear, we lack an understanding of how early redox signaling is coordinated temporally and spatially to mediate long-term tissue repair. We recently reported that burn injury induces a distinct repair response with impaired collagen remodeling and delayed healing (LeBert et al., 2018; Miskolci et al., 2019). In light of the known defect in sensory function after burn injuries in humans, we sought to determine how the early epithelial response modulates sensory axon recovery. Our findings suggest that early damage-induced keratinocyte movement plays a role in the spatial patterning of ROS production in the wound microenvironment and impacts the ability of sensory axons to regenerate.

Collective keratinocyte migration is conserved across species and is required to mediate wound closure after tissue injury (Kirfel and Herzog, 2004; Mayor and Etienne-Manneville, 2016). While collective cell migration has been observed in larval zebrafish previously (Olson and Nechiporuk, 2021; Yamaguchi et al., 2022), its contribution to tissue repair remains unclear. In comparison to the organized movement associated with keratinocyte response to mechanical injury, our observations here identify excessive keratinocyte migration as a defining feature of the response to burn injury. Basal keratinocyte migration following burn injury appeared to lack a stereotypical leader-follower dynamic, with cells instead moving independently of one another but as a collective group. This observation suggests that collective keratinocyte migration is a feature of tissue repair in larval zebrafish regardless of the mode of injury and that its regulation is required for the success of long-term healing. Indeed, we provide evidence that early migration and formation of lamellipodia in basal keratinocytes requires Arp2/3 signaling, and that this aberrant migration regulates the temporal and spatial distribution of early redox signaling in the wound tissue. However, Arp2/3 inhibition did not lead to sustained control of keratinocyte migration and production of ROS eventually increased throughout the tail fin. Accordingly, sensory function was not significantly improved at 24 hours post wound.

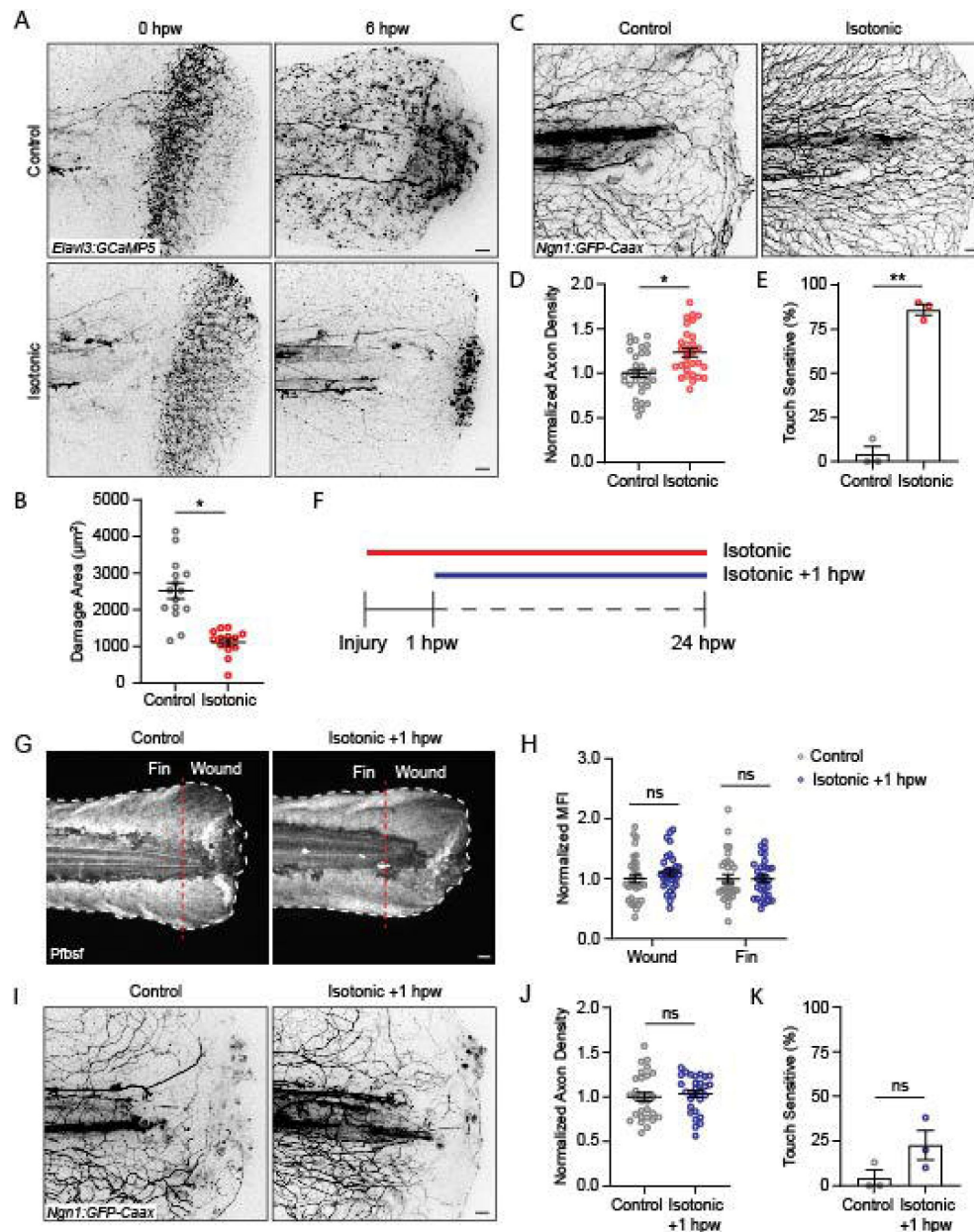


Figure 6

Isotonic treatment improves axon regeneration.

(A) Confocal max-projected images of axon damage in control or isotonic treated *Tg(Elavl3:GCaMP5)* larvae 0 or 6 hpw. (B) Quantification of axon damage in control and isotonic treated burned fins at 6 hpw as treated in A. N=16 larvae per condition from 3 replicates. (C) Confocal max-projected images of sensory axons in larvae 24 hpw as treated in A. (D) Quantification of axon density 24 hpw in larvae treated as depicted in C. N>30 larvae per condition from 3 replicates. (E) Quantification of sensory perception 24 hpw in larvae treated as in C. N=24 larvae each from 3 replicates. (F) Schematic illustrating the different isotonic treatment paradigms that are being compared. (G) Confocal sum-projected images of pfbsf intensity in control and isotonic +1 hpw treated burned larvae. Dashed red line denotes the boundary between the wound area and distal fin tissue. White dashed line denotes the fin. (H) Quantification of mean Pfbsf fluorescence intensity (MFI) 6 hpw in the indicated region of the fin normalized to the control condition. N=31 larvae per condition from 3 replicates. (I) Confocal max-projected images of sensory axons 24 hpw in burned control or isotonic treated larvae starting 1 hpw. (J) Quantification of axon density 24 hpw in larvae treated as in D. N=29 larvae per condition from 3 replicates. (K) Quantification of sensory perception 24 hpw in larvae treated as in D. N=24 larvae per condition from 3 replicates. Unless otherwise indicated, scale bars=20 μ m. * p <0.05, ** p <0.01, ns=not significant.

To further modulate keratinocyte movement, we took advantage of isotonic treatment that is known to affect keratinocyte migration after wounding by tail fin transection. Cell swelling is thought to induce migration by promoting branched actin polymerization and lamellipodia formation (Bera et al., 2022 [↗](#); Han et al., 2012 [↗](#); Sforza et al., 2022 [↗](#)), potentially through the activity of mechanically activated ion channels. Previous groups have shown that osmotic differences trigger both ATP release and lamellipodia formation in basal keratinocytes, which promote keratinocyte migration (Chen et al., 2019 [↗](#); Gault et al., 2014 [↗](#)). Our results also show that osmotic modulation affects basal keratinocyte motility in response to burn injury. Interestingly, limiting keratinocyte motility by isotonic treatment is detrimental to tissue repair following mechanical injury (Gault et al., 2014 [↗](#)), but isotonic treatment both rescues epithelial morphology and reduces axon damage following thermal injury. This suggests that there is an optimal amount of keratinocyte movement needed for efficient repair and long-term regeneration.

A conceptual challenge in wound repair has been understanding how early wound-induced events are linked to long-term repair (Sonnemann and Bement, 2011 [↗](#)). ROS signaling provides a framework to understand this link due to its requirement for both early wound contraction and long-term regeneration. In zebrafish, the reactive oxygen species H₂O₂ is generated along a tissue-scale gradient with the highest levels at the wound edge (Niethammer et al., 2009 [↗](#)). While this spatial gradient undoubtedly directs cell function based on the position along the gradient, uncontrolled ROS production is damaging to tissues. Therefore, a mechanism must exist to control ROS such that it remains relatively localized to the site of damage and is controlled temporally and spatially. In addition to their early role in wound resealing, keratinocyte redox signaling is critical for long-term repair. Cell swelling induces cPLA₂-dependent 5-oxoETE production and immune cell recruitment (Enyedi et al., 2013 [↗](#)). This suggests the signals that control keratinocyte motility may simultaneously modify long-term keratinocyte signaling.

Our findings suggest that cell migration can modulate tissue scale signaling following injury. Importantly, isotonic treatment blocks keratinocyte movement and restores localized ROS signaling at the wound edge. The dependence on migration was supported both by the effects of Arp2/3 inhibition on early ROS signaling, and by the finding that isotonic treatment started after migration was complete (1 hour after injury) did not rescue tissue scale ROS. This suggests that early keratinocyte migration patterns the tissue scale ROS distribution. Additionally, ROS inhibition with the drug DPI provided only slight benefits to axon regeneration. This could be due to treatment being sustained for only 1 h, but migration was also not inhibited which further points to the importance of migration in early wound signaling. The finding that isotonic treatment at the time of injury was sufficient to rescue sensory function but treatment after 1 hour did not rescue axon regeneration further highlights the importance of this early motile response for setting up the longer-term repair after damage.

The benefit of a system in which keratinocyte motility controls downstream signaling is two-fold. First, it enables signaling to be scaled to the size of injury. If more cells migrate due to a larger injury, then production of ROS will likewise increase. Second, this system provides a mechanism to control the spatial localization of signaling. Keratinocyte migration requires transiently detaching from neighboring cells. Thus, the act of migrating induces a physical change in the tissue that demarcates the wound region from healthy tissue. It is known that production of ROS promotes keratinocyte motility, and that adhesion is linked to cellular redox state (Dunnill et al., 2015 [↗](#); Mendieta-Serrano et al., 2019 [↗](#)). Given these observations, it seems plausible that ROS production in wounded tissue is linked to the biomechanical state of keratinocytes – with low ROS in static, adhered cells, and high ROS in loosely adhered or migrating cells. A conceptual framework such as this would explain excessive ROS production in burn wounded tissue. Early keratinocyte dynamics in burned tissue are associated with normal wound edge ROS production. However, lack of a migratory stop signal may result in excessive keratinocyte migration and subsequent epithelial damage associated with keratinocytes detaching from the basal lamina. Therefore, failure to restore epithelial homeostasis due to unabated keratinocyte movement may allow for ROS

production to continue over time and spread further away from the wound site. Future studies will be aimed at identifying the molecular link between cell migration and tissue scale signaling during tissue repair.

In summary, we have identified early wound-induced keratinocyte migration as a mechanism that controls spatial patterning of long-term wound signaling. These findings highlight the ability of keratinocytes within the wound microenvironment to integrate early signaling and migratory functions that mediate initial wound closure and subsequently regulate spatial tissue signaling necessary for efficient repair of sensory neuron function. Further, our results not only highlight the utility of larval zebrafish for revealing new insights of the tissue response to injury *in vivo*, but also demonstrate the potential for these findings to inform new treatment strategies for wound healing more broadly.

Materials & Methods

Ethics

This study was carried out in accordance with the recommendations from the Guide for the Care and Use of Laboratory Animals from the National Institutes of Health. All zebrafish protocols in this study were approved by the University of Wisconsin-Madison Research Animals Resource Center (Protocol M005405-R02).

Zebrafish maintenance and handling

Adult zebrafish and embryos were maintained as described previously (Houseright et al., 2021 [DOI](#); Miskolci et al., 2019 [DOI](#)). For all experiments, 3 days post-fertilization (dpf) larvae were anesthetized in E3 medium containing 0.2 mg/mL Tricaine (ethyl 3-aminobenzoate; Sigma-Aldrich) and maintained at 28.5 °C. All transgenic lines including *Tg(Ngn1:GFP-Caax)* (Blader et al., 2003 [DOI](#)), *Tg(Krt4:LifeAct-mRuby)* (Lam et al., 2015 [DOI](#)), *Tg(Krt4:UtrCH-GFP)*, *Tg(Krt4:TdTomato)*, *TgBac(Lamc1:Lamc1-sfGFP)* (Yamaguchi et al., 2022 [DOI](#)), *Tg(ElavI3:GCaMP5)* (Akerboom et al., 2012 [DOI](#)), *Tg(Krtt1c19e:LifeAct-mRuby)*, and *Tg(Krtt1c19e:acGFP)* were maintained on the AB background strain. To screen larvae for fluorescence, a Zeiss Zoomscope EMS3/SyCoP3 with a Plan-Neofluar Z objective was used.

Generation of *Tg(Krtt1c19e:LifeAct-mRuby)* transgenic line

The *Krtt1c19e* promoter (Lee et al., 2014 [DOI](#)) flanked by Age1 and NotI was isolated and cloned into an expression vector containing Lifeact-mRuby and Tol2 elements for genomic integration. 3 nL of solution made of 50 ng DNA and 25 ng Tol2 transposase mRNA were injected into the yolk of one-cell stage embryos. F0 larvae were raised to adulthood and crossed to adult AB zebrafish. F2 larvae were screened for mRuby expression and grown to generate stable lines. *Tg(Ngn1:GFP-Caax)* larvae transiently expressing *Krtt1c19e:Lifeact-mRuby* were used for simultaneous imaging of axon-keratinocyte interactions, and larvae transiently expressing *Krtt1c19e:Lifeact-mRuby* were used to acquire timelapses of basal keratinocyte migration in the CK666 treatment condition.

Caudal fin transection and burn injury

All injuries were applied to fish anesthetized in 1x Tricaine with E3. Transection of the caudal fin was performed on anesthetized larvae in a 60 mm tissue culture-treated dish containing 1x Tricaine with E3. Larvae were cut perpendicular to the caudal notochord boundary using a surgical blade (Feather No. 10, VWR). Burn injury was performed on anesthetized larvae in a 60 mm tissue culture-treated dish containing 1x Tricaine with E3. A fine tip cautery pen (Geiger Medical Technologies) was used to burn the caudal fin until the wounded region reached halfway to the posterior notochord boundary.

After injury, larvae were kept in 60 mm dishes and maintained at 28.5 C until imaged. For two-wound experiments, larvae were either transected or burned as described above. Secondary transection, after either 5 minutes or 6 hours, was performed as described above.

Drug treatment

For all treatments, larvae were incubated in the indicated drug solution for at least 15-30 minutes. Each drug solution was made containing 1x Tricaine with E3 to keep larvae anesthetized during the experiment. Unless indicated otherwise, larvae were in the presence of treatment for the duration of all experiments. All treatments did not obviously impair larval development or health, and axon density in unwounded larvae was measured for each treatment to ensure there were no deficits in axonal patterning (Figure Supplement 3D,E). To elicit sensory axon damage in the absence of a wound, larvae were treated with 1.5% sodium azide (Fisher Scientific). Isotonic medium was prepared by supplementing 1x Tricaine with E3 with either NaCl (Fisher Scientific) or D-Sorbitol (Sigma-Aldrich) to a final concentration of 135 mM. Isotonic medium does not noticeably impair embryonic development or health of the larvae on the time scale used here (Gault et al., 2014 [DOI](#); Jelcic et al., 2019 [DOI](#)). For experiments using CK666 (Sigma), larvae were incubated in 100 μ M Ck666 for one hour before wounding and kept in treatment until 6 hpw. The DPI dosage of 20 μ M used in this manuscript was determined by dosage curve of 1 μ M to 200 μ M. Larvae were pre-treated for 1 hour and then burned and incubated in treatment for 1 hour - treatments over 20 μ M or over 1 hpw were detrimental to larval health and development.

Live and time-lapse imaging

Larvae were imaged using a spinning disc microscope (CSU-X, Yokogawa) with a confocal scanhead on a Zeiss Observer Z.1 inverted microscope, Plan-Apochromat NA 0.8/20X objective, and a Photometrics Evolve EMCCD camera. All images were acquired using ZEN 2.6 software. For time lapse imaging, larvae were mounted in a zWEDGI restraining device (Huemmer et al., 2016 [DOI](#)) with the head covered in 2% low-melting point agarose (Sigma-Aldrich). For single time point imaging, anesthetized larvae were mounted in 2% low-melting point agarose on a 35 mm glass-bottom depression dish (CellVis). In all cases, larvae were imaged in E3 medium supplemented with Tricaine as described above.

Quantification of axon density and sensory function

Axon density was measured by generating maximum intensity z-projected confocal images of the caudal fin using FIJI (Schneider et al., 2012 [DOI](#)). For every experiment, the caudal fin area posterior to the notochord was outlined using the Polygon tool and measured to obtain a total surface area ROI. Axons inside the outlined area were manually thresholded so all axons posterior to the notochord were labeled and no saturated pixels were present. Density was measured by dividing the area of detected axons by the area of the ROI. In each case, density values of the experimental sample were normalized to the indicated control - either unwounded or control treated fins. Sensory neuron function was determined using a behavioral touch assay (Granato et al., 1996 [DOI](#)). 3-day post-fertilization larvae were wounded as described above. At the indicated time post-wound, larvae were briefly anesthetized for mounting into the zWEDGI restraining device, with only the head mounted in 2% low-melting point agarose. Fresh E3 was added, and larvae were allowed to rest for one hour.

To assess sensory function, the wounded region of caudal fin was touched with the tip of an eyelash brush (No. 1 Superfine Eyelash, Ted Pella) and the presence or absence of a twitch reflex was recorded.

Quantification of axon damage

Maximum intensity z-projected confocal images of the caudal fin were generated using FIJI. For all experiments, the caudal fin area posterior to the notochord was outlined using the Polygon tool and measured to obtain a total surface area ROI. Axon fragments inside the outlined area were manually thresholded so all fragments posterior to the notochord were labeled and no saturated pixels were present, and an area measurement of these thresholded pixels was taken.

Visualization of sensory axon and tissue damage

To visualize damage to sensory axons, *Tg(Elavl3:GCaMP5)* larvae were used. Identical microscope settings (20x objective, 10% laser, 100 ms exposure, 2 μ m step size) were used for all experiments to acquire images and movies. Representative images are maximum intensity z-projections of the caudal fin generated using FIJI. FM 1-43 dye (Life Technologies) was used to visualize tissue damage following transection and burn injury.

For these experiments, larvae were incubated in 1 mg/ml FM 1-43 for 15 minutes prior to injury and through time at which they were imaged. Larvae were maintained at 28.5 C until imaging at the indicated time post-injury.

Quantification of hydrogen peroxide level

Hydrogen peroxide was quantified using pentafluorobenzenesulfonyl fluorescein (Pfbfs, Santa Cruz) (Maeda et al., 2004 [\[link\]](#)). Larvae were incubated in 1 μ M Pfbfs for 15 minutes prior to injury and maintained in dye solution for the duration of each experiment. Identical microscope settings (10x objective, 1% laser, 50 ms exposure, 3.7 μ m step size) were used for all experiments to acquire images and movies. Pfbfs intensity was calculated by generating sum projections and measuring mean gray value of the fin and wound zone in FIJI. Wound zone Pfbfs quantifications were taken by measuring mean gray value of the area posterior to the notochord. For measurements of the fin, mean gray value of the trunk area 200 μ m anterior to the tip of the notochord and excluding pigmented skin within the region was measured. Background signal was subtracted for each measurement.

Cell tracking

Basal keratinocyte tracking following tissue injury was performed using *Tg(Krtt1c19e:acGFP)* larvae. Cell tracking was performed using the Spots module in Imaris version 9.8.2 (Bitplane, Zurich, Switzerland). For each larva, 3 representative cells were identified and manual tracking was performed, with the average of these cells being used to generate a single value for further analysis. To control for drift of the entire fin during imaging, non-moving pigment was manually tracked by Brightfield and track length was subtracted from basal keratinocyte movement. In all cases, larvae were imaged for 1 hour following injury at an interval of 30 seconds.

Morpholino injection

Ngn1 morpholino with the sequence 5'-ACG ATC TCC ATT GTT GAT AAC CTG G-3' (Cornell and Eisen, 2002 [\[link\]](#)) was used to prevent sensory neuron formation in *Elavl3-GCaMP5* larvae to confirm damage signals were constrained only to axons. 5 ng of Ngn1 morpholino was injected into the yolk of one-to two-cell stage zebrafish embryos. Larvae were incubated at 28.5 C until used for experiments at 3 days post-fertilization. Before use in experiments, larvae were screened by the sensory function assay described above to ensure that sensory neurons were depleted.

Image processing

Images were processed and analyzed using FIJI and Imaris version 9.8.2 as indicated. Supplemental movies were generated in FIJI and edited using Adobe Premiere Pro (Adobe). In Adobe Premiere Pro, pseudocoloring of individual keratinocytes was done using the Color Effects module with manual tracking.

Statistical analysis

Each experimental condition consists of at least three independent biological replicates, defined as three clutches of larvae spawned on three different days. Cell-tracking experiments were analyzed using non-parametric methods (Wilcoxon rank-sum test). Quantification of axon density was analyzed using linear mixed-effect models in which biological replicate was treated as a random effect and experimental conditions (e.g., wound, time, or chemical treatment) treated as fixed factors. Experiments measuring fluorescence intensity (Pfbf intensity) were analyzed in the same manner, except the response (fluorescence) was log-transformed prior to analysis. Means computed on the log scale estimate the median response when back transformed to original units. At the same time, differences between means (of log-transformed data) become ratios of medians after back transformation to the original scale (Aitkin M et al., 1989 [DOI](#)). Experiments involving the proportion of fish that responded to touch were analyzed using a general linear model that included replicate and experimental condition as fixed effects; standard errors used for estimation and testing were adjusted to correct for heteroscedasticity in the proportions (Long and Ervin, 2000 [DOI](#)). Graphing was performed using GraphPad Prism 9 (GraphPad Software, Inc, San Diego, CA). Sample size is reported for specific experiments in the Figure legends.

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Additional information

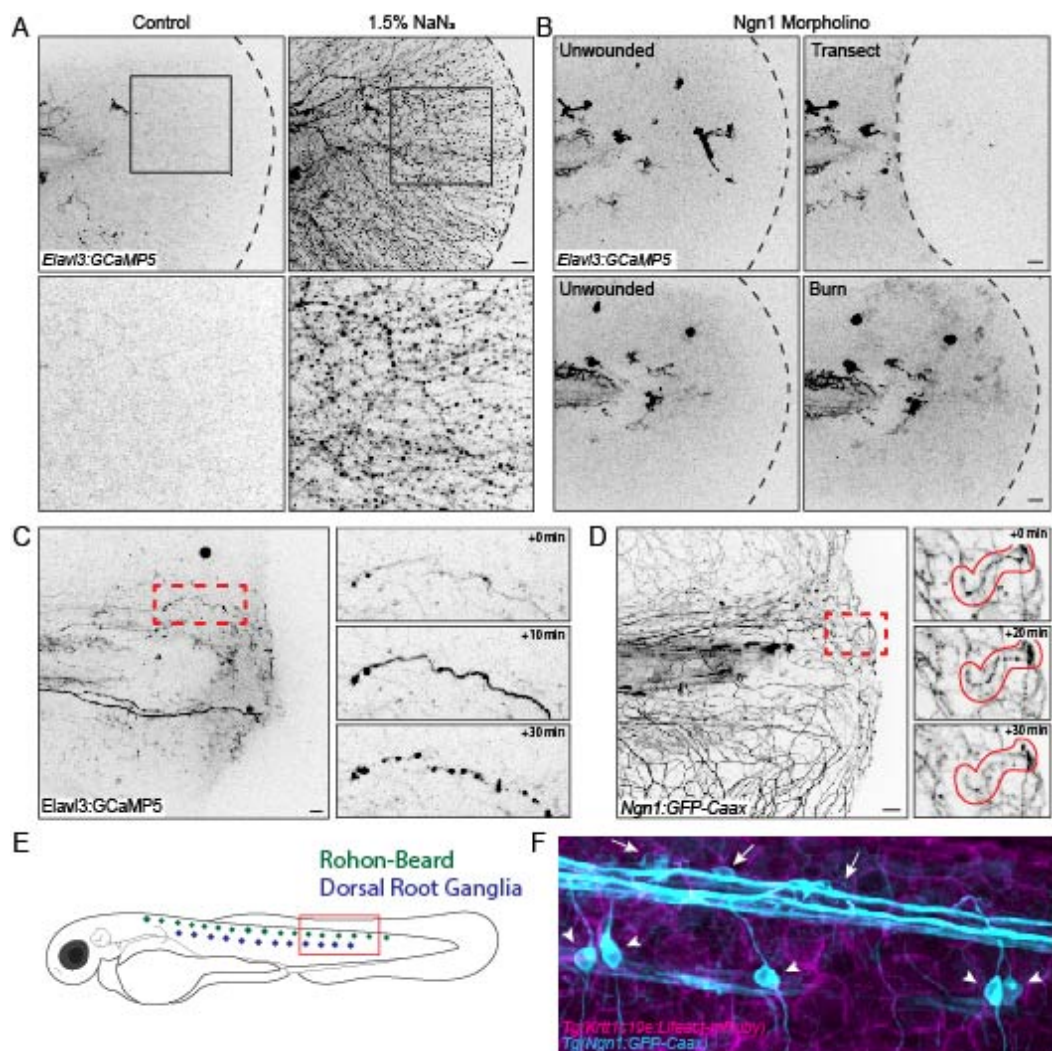
Author Contributions

1. Fister, A. Horn, and A. Huttenlocher designed and planned the experimental approach. A. Fister and A. Horn performed all the experiments. M. Lasarev performed all statistical analysis. A. Fister, A. Horn, and A. Huttenlocher wrote the paper.

Declaration of Interests

The authors declare no competing financial interests.

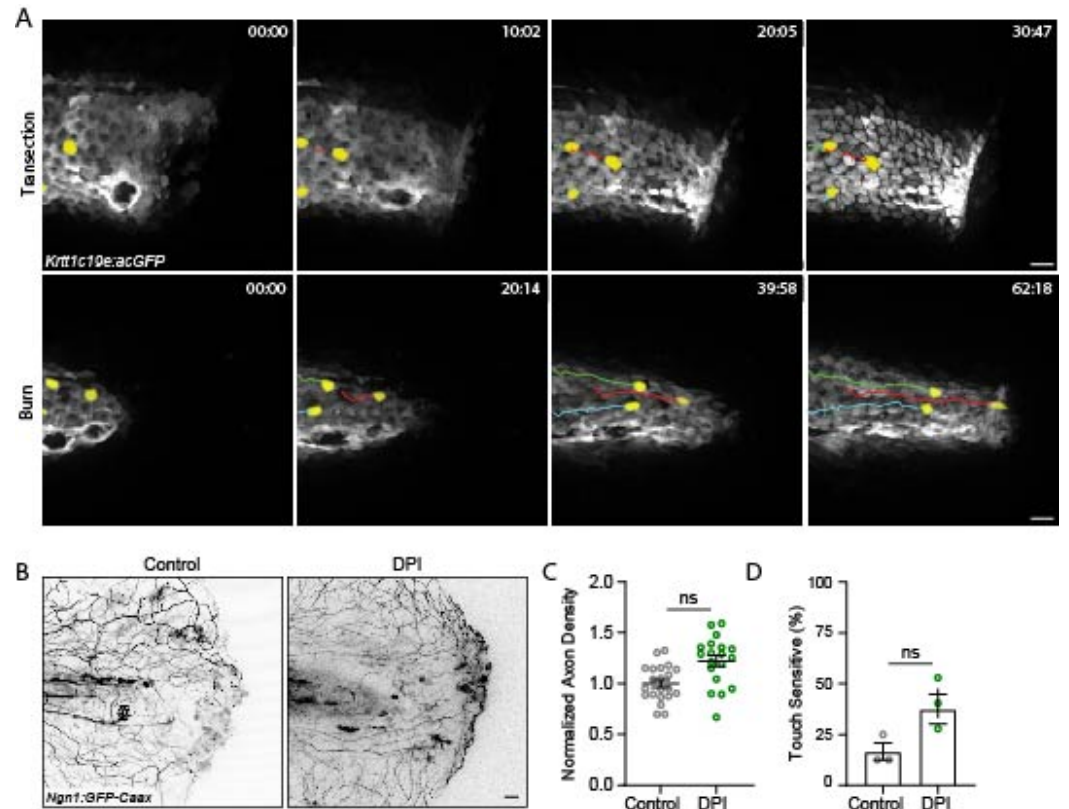
Supplemental Figures



Supplemental Figure 1

Elavl3-GCaMP5 transgenic fish show sensory axon damage.

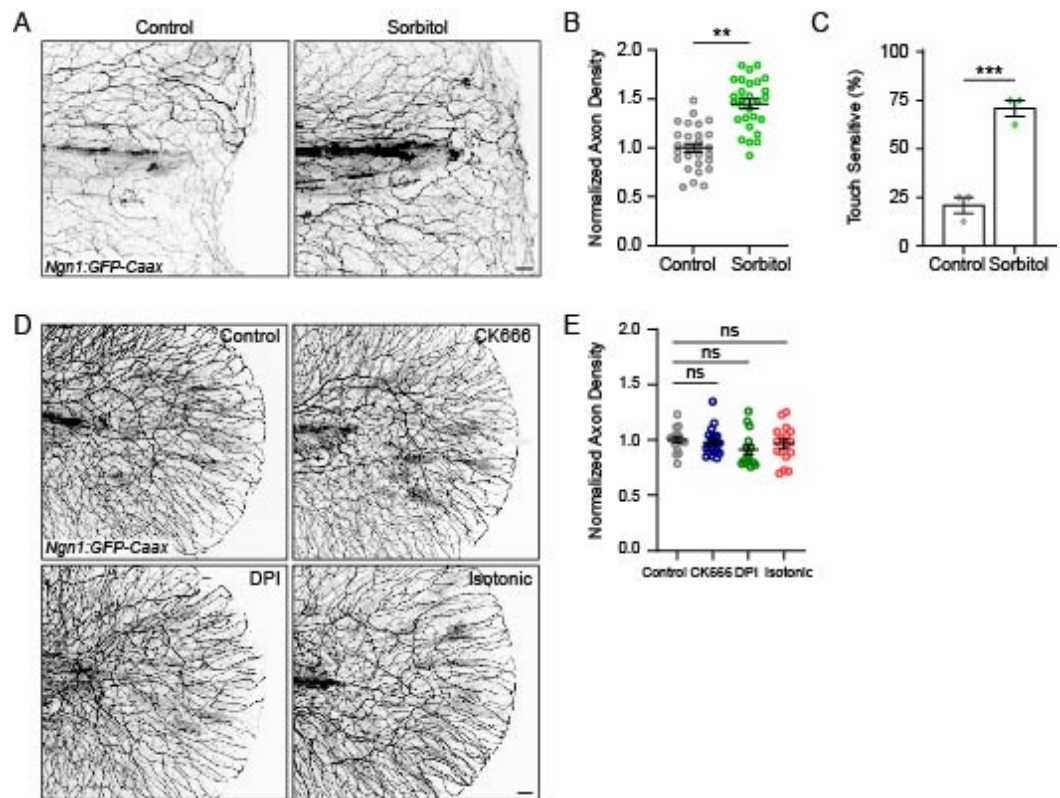
(A) Confocal max-projected images of axon damage in *Tg(Elavl3:GCaMP5)* larval zebrafish caudal fins either untreated or 30 minutes post-treatment with the neurotoxin sodium azide (NaN_3 , 1.5% final concentration). Sensory neuron damage is indicated by calcium-positive axon fragments (black dots). Dashed black lines denote the fin edge. Black boxes highlight area of inset, shown below. (B) Confocal max-projected images of *Tg(Elavl3:GCaMP5)* larvae injected with Ngn1 morpholino both before and 5 minutes after the indicated injury. (C) Confocal max-projected images of *Tg(Elavl3:GCaMP5)* larvae taken from a time series. Red dashed box denotes inset area shown on right of a sensory axon fragmenting over a period of 30 minutes. (D) Confocal max-projected images of *Tg(Ngn1:GFP-Caax)* larvae taken from a time series. Red dashed box denotes inset area shown on right of a sensory axon fragmenting over a period of 30 minutes. (E) Schematic of Rohon-Beard (green) and Dorsal Root Ganglia (blue) soma localization in 3 dpf zebrafish. Red box denotes area in which the image shown in F was acquired. (F) Representative confocal max-projected image of intact RB and DRG somas 24 hpw in a *Tg(Ngn1:GFP-Caax); Tg(Krtt1c19e:Lifeact-mRuby)* dual-labeled larva. Arrows denote RB somas, while arrowheads indicate DRG somas. Soma position was unchanged compared to pre-wounding. In all cases, scale bar=20 μm .



Supplemental Figure 2

Inhibition of early ROS signaling does not significantly improve sensory neuron function.

(A) Confocal time series of basal keratinocyte, *Tg(Krtt1c19e:acGFP)*, movement after the indicated injury. Yellow pseudocolored cells highlight keratinocyte displacement. (B) Confocal max-projected images of sensory axons treated with DPI. (C) Quantification of axon density 24 hpw in larvae treated as stated in F. N>19 larvae per condition from 3 replicates. (D) Quantification of sensory perception 24 hpw in larvae treated as in F. N=24 larvae each from 3 replicates. Unless otherwise indicated, scale bars=20 μ M. ns=not significant.



Supplemental Figure 3

Keratinocyte movement after injury and effect of D-Sorbitol on sensory axon regeneration.

(A) Representative max-projected confocal images of sensory axons in control and isotonic D-Sorbitol treated larvae 24 hpw. (B) Quantification of axon density in wounded tissue 24 hpw. N=28 larvae each from 4 replicates. (C) Quantification of sensory perception 24 hpw. N=24 larvae each collected from 4 replicates. (D) Representative max-projected confocal images of sensory axons in unwounded, drug-treated 4 dpf larvae. (E) Quantification of axon density in larvae treated as stated in E. N>15 larvae each from 3 replicates. In all cases, scale bar=20 μ m. **p<0.01, ***p<0.001, ns=not significant.

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Reviewer #1 (Public Review):

Summary:

In this manuscript, Fister et. al. investigate how amputational and burn wounds affect sensory axonal damage and regeneration in a zebrafish model system. The authors discovered that burn injury results in increased peripheral axon damage and impaired regeneration. Convincing experiments show altered axonal morphology and increased Ca²⁺ fluxes as a result of burn damage. Further experimental proof supports that early removal of the burnt tissue by amputation rescues axonal damage. Burn damage was also shown to markedly increase keratinocyte migration and increase localized ROS production as

measured by the dye Pfbf. These responses could be inhibited by Arp 2/3 inhibition and isotonic treatment.

Strengths:

The authors use state-of-the-art methods to study and compare transection and burn-induced tissue damage. Multiple experimental approaches (morphology, Ca^{2+} fluxing, cell membrane labeling) confirm axonal damage and the impaired regeneration time. Furthermore, the results are also accompanied by functional response tests of touch sensitivity. This is the first study to extend the role of tissue-damage related osmotic exposure beyond wound closure and leukocyte migration to a novel layer of pathology: axonal damage and regeneration.

The authors provide elegant experiments showing that early removal of the burnt tissue can rescue damage-induced axonal damage, which could also be interpreted in an osmotic manner. In the revised version of the paper the authors indeed show that tail fin transections close faster than burn wounds, allowing for lower hypotonic exposure time. However, their new experiments suggest that axonal damage and slow regeneration in tail fin burn wounds are not a direct consequence of the extended exposure time to hypotonic water.

Weaknesses:

The conclusions of the paper claiming a link between burn-induced epithelial cell migration, spatial redox signaling, and sensory axon regeneration are mainly based on correlative observations. Arp 2/3 inhibition impairs cell migration but has no significant effect on axon regeneration and restoration of touch sensitivity.

Genetic approaches have been tested during the revision process to directly prove the role of ROS production by targeting DUOX, however, the combination of DUOX morpholino and burn injury was lethal to the larvae and long-term pharmacological inhibition over 1 hour was also detrimental.

<https://doi.org/10.7554/eLife.94995.2.sa2>

Reviewer #3 (Public Review):

Fister and colleagues use regeneration of the larval zebrafish caudal fin to compare the effects of two modes of tissue damage-transection and burn-on cutaneous sensory axon regeneration. The authors found that restoration of sensory axon density and function is delayed following burn injury compared to transection.

The authors hypothesized that thermal injury triggers signals within the wound microenvironment that impair sensory neuron regeneration. The authors identify differences in the responses of epithelial keratinocytes to the two modes of injury: keratinocytes migrate in response to burn but not transection. Inhibiting keratinocyte migration with a small-molecule inhibitor of Arp2/3 (CK666) resulted in decreased production of reactive oxygen species (ROS) at early, but not late, timepoints. Preventing keratinocyte migration by wounding in isotonic media resulted in increased sensory function 24 hours after burn.

Strengths of the study include the beautiful imaging and rigorous statistical approaches used by the authors. The ability to assess both axon density and axon function during regeneration is quite powerful. The touch assay adds a unique component to the paper and strengthens the argument that burns are more damaging to sensory structures and that different treatments help to ameliorate this.

A weakness of the study is the lack of genetic and cell autonomous manipulations. Additional comparisons between transection and burns, in particular with manipulations that specifically modulate ROS generation or cell migration without potentially confounding effects on other cell types or processes would help to strengthen the manuscript. In terms of framing their results, the authors refer to "sensory neurons" and "sensory axons" throughout the text - it should be made clear what type of neuron(s)/axon(s) are being visualized/assayed. Along these lines, a broader discussion of how burn injuries affect sensory function in other systems-and how the authors' results might inform our understanding of these injury responses-would be beneficial to the reader.

In summary, the authors have established a tractable vertebrate system to investigate different sensory axon wound healing outcomes in vivo that may ultimately allow for the identification of improved treatment strategies for human burn patients. Although the study implicates differences in keratinocyte migration and associated ROS production in sensory axon wound healing outcomes, the links between these processes could be more rigorously established.

<https://doi.org/10.7554/eLife.94995.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

In this manuscript, Fister et. al. investigate how amputational and burn wounds affect sensory axonal damage and regeneration in a zebrafish model system. The authors discovered that burn injury results in increased peripheral axon damage and impaired regeneration. Convincing experiments show altered axonal morphology and increased Ca²⁺ fluxes as a result of burn damage. Further experimental proof supports that early removal of the burnt tissue by amputation rescues axonal damage. Burn damage was also shown to markedly increase keratinocyte migration and increase localized ROS production as measured by the dye Pfbfsf. These responses could be inhibited by Arp 2/3 inhibition and isotonic treatment.

Strengths:

The authors use state-of-the-art methods to study and compare transection and burn-induced tissue damage. Multiple experimental approaches (morphology, Ca²⁺ fluxing, cell membrane labeling) confirm axonal damage and impaired regeneration time. Furthermore, the results are also accompanied by functional response tests of touch sensitivity. This is the first study to extend the role of tissue-damage-related osmotic exposure beyond wound closure and leukocyte migration to a novel layer of pathology: axonal damage and regeneration.

Weaknesses:

The conclusions of the paper claiming a link between burn-induced epithelial cell migration, spatial redox signaling, and sensory axon regeneration are mainly based on correlative observations. Arp 2/3 inhibition impairs cell migration but has no significant effect on axon regeneration and restoration of touch sensitivity.

We agree with the reviewer. We have tried many experiments to address this question. The data show that Arp 2/3 inhibition with CK666 is an effective way to inhibit initial keratinocyte migration. However, later migration still proceeds. What is interesting is that just inhibition of the early migration is sufficient to restore localized ROS production in the wound area in the first hour post-burn, even if this is not sufficient to prevent ROS accumulation over time. There is also a trend toward improved sensory neuron function late after this early treatment. However, this is not statistically significant. We think it is likely that both migration and tissue scale ROS influence the regeneration defect of sensory neurons after burn. The data using isotonic solution supports this conclusion. We have tried many other ways to limit keratinocyte migration including depletion of talin and expression of a dominant negative Rac in basal epithelial cells, but these treatments were not compatible with survival of the fish after burn.

Pharmacological or genetic approaches should be used to prove the role of ROS production by directly targeting the known H2O2 source in the system: DUOX.

We agree that pharmacologic or genetic approaches to directly manipulate ROS production would provide substantial support to the hypothesis that ROS, along with keratinocyte migration, is a main factor contributing to poor burn outcomes. To address this, we first tried using a morpholino to deplete DUOX. However, the combination of DUOX morpholino and burn injury was lethal to larvae. We also used pharmacologic inhibition of ROS production using DPI (Diphenyleneiodonium). With this treatment, ROS is inhibited for only the first hour post-burn as treatment is lethal for longer periods of time. Burned larvae have marginally improved axon density and touch sensitivity, suggesting the importance of ROS in burn outcomes, however it was not statistically significant. It is likely that an increased effect would be observed with longer treatment, but treatment for more than 1 hour was toxic. We have added a supplemental figure with this new DPI data.

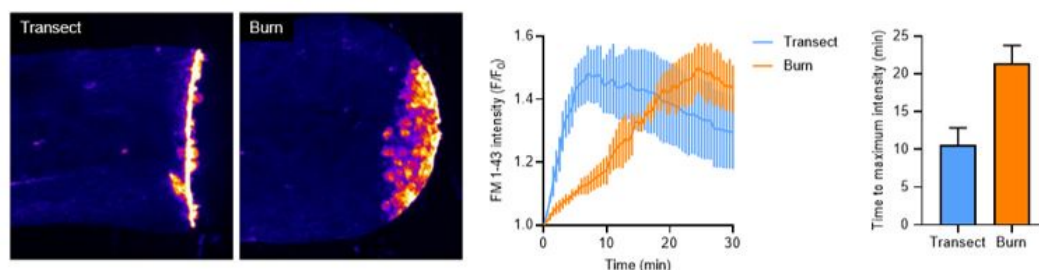
While the authors provide clear and compelling proof that osmotic responses lie at the heart of the burn-induced axonal damage responses, they did not consider the option of further exploring any biology related to osmotic cell swelling. Could osmotic ATP release maybe play a role through excitotoxicity? Could cPLA2 activation-dependent eicosanoid production relate to the process? Pharmacological tests using purinergic receptor inhibition or blockage of eicosanoid production could answer these questions.

We agree that the role of osmotic cell swelling in the burn response is an interesting avenue for future study. However, we make use of isotonic treatment in this study specifically for its effect on keratinocyte migration and broad-scale wound healing. As a result, we feel that pursuing the biology of this swelling phenomenon is outside the scope of this paper.

The authors provide elegant experiments showing that early removal of the burnt tissue can rescue damage-induced axonal damage, which could also be interpreted in an osmotic manner: tail fin transections could close faster than burn wounds, allowing for lower hypotonic exposure time. Axonal damage and slow regeneration in tail fin burn wounds could be a direct consequence of extended exposure time to hypotonic water.

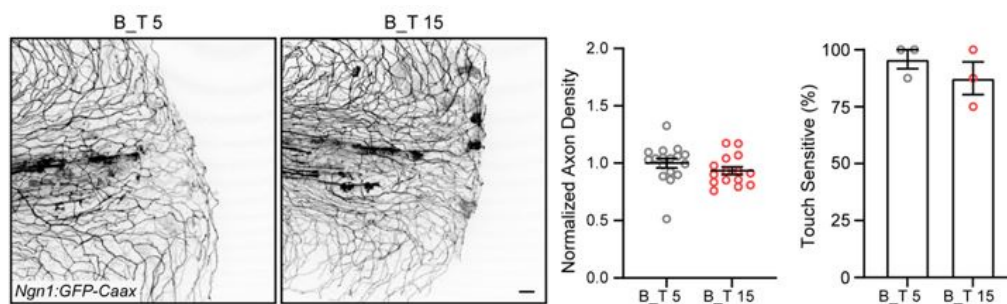
We have done experiments using FM dye to test how long it takes burn and transection wounds to close (shown below). In these experiments, dye entry into wounded tissue is used as a readout of wound closure. Dye is only able to enter wounded tissue when the epithelial barrier is disrupted. Our data reveal that transections take approximately 10 minutes to fully close, while burns take approximately 20 minutes to close.

Author response image 1.



To test if this difference in wound closure time would have an effect on axon outcomes, we repeated, but slightly modified, the dual-wound experiment. We increased the amount of time the burn condition was exposed to hypotonic conditions by 10 additional minutes (by transecting burned tissue at 15 minutes post burn, shortly before closure) and compared axon outcomes to the 5 mpw control transection. These results show there was no difference in axon regeneration or function when secondary transection was performed at 5 or 15 minutes post burn, suggesting that increased exposure to hypotonic solution is not the reason for defects in axon outcomes after burn injury.

Author response image 2.



Reviewer #2 (Public Review):

This is an interesting study in which the authors show that a thermal injury leads to extensive sensory axon damage and impaired regrowth compared to a mechanical transection injury. This correlates with increased keratinocyte migration. That migration is inhibited by CK666 drug treatment and isotonic medium. Both restrict ROS signalling to the wound edge. In addition, the isotonic medium also rescues the regrowth of sensory axons and recovery of sensory function. The findings may have implications for understanding non-optimal re-innervation of burn wounds in mammals.

The interpretation of results is generally cautious and controls are robust.

Here are some suggestions for additional discussion:

The study compares burn injury which produces a diffuse injury to a mechanical cut injury which produces focal damage. It would help the reader to give a definition of wound edge in the burn situation. Is the thermally injured tissue completely dead and is resorbed or do axons have to grow into damaged tissue? The two-cut model suggests the

latter. Also giving timescales would help, e.g. when do axons grow in relation to keratinocyte movement? An introductory cartoon might help.

We thank the reviewer for these insightful comments and questions. The burn wound is defined as the area that is directly damaged as a result of increased heat (labeled by FM dye entry), and the burn wound edge as the first line of healthy cells adjacent to the burned cells. These definitions have been added to the text to clarify the areas referenced. Recent experiments lead us to believe the wound area is composed almost completely of dead cells, but we are currently working to discover the fate of these dead cells as well as the wound adjacent cells that migrate to the wound edge after burn. As a result, we do not know whether axons grow into damaged tissue or if the damaged tissue is extruded, but we do see growth cone formation within a few hours after wounding suggesting the axons are actively trying to regenerate after a burn.

Could treatment with CK666 or isotonic solution influence sensory axons directly, or through other non-keratinocyte cell types, such as immune cells?

We have done experiments looking at the density of caudal fin innervation in CK666, isotonic, or DPI treated fins. The axon density is unchanged in all these treatments compared to control treated larvae, so we do not believe these treatments affect axon health homeostatically. These data have been added to supplemental figure 3. Additionally, one of the benefits of the larval zebrafish burn model is the simplicity of the system – the epidermis is primarily composed of sensory axons, mesenchymal cells and keratinocytes. The burn environment is proinflammatory so it does promote immune cell recruitment, but we do not believe the immune cells are interacting directly with sensory axons besides clearing axonal debris. Previous papers by our lab have shown that peak immune cell recruitment occurs at 6 hpw, but they localize to the damaged tissue in the burn area and not the wound edge.

Reviewer #3 (Public Review):

Fister and colleagues use regeneration of the larval zebrafish caudal fin to compare the effects of two modes of tissue damage-transection and burn-on cutaneous sensory axon regeneration. The authors found that restoration of sensory axon density and function is delayed following burn injury compared to transection.

The authors hypothesized that thermal injury triggers signals within the wound microenvironment that impair sensory neuron regeneration. The authors identify differences in the responses of epithelial keratinocytes to the two modes of injury: keratinocytes migrate in response to burn but not transection. Inhibiting keratinocyte migration with the small-molecule inhibitor of Arp2/3 (CK666) resulted in decreased production of reactive oxygen species (ROS) at early, but not late, time points. Preventing keratinocyte migration by wounding in isotonic media resulted in increased sensory function 24 hours after burn.

Strengths of the study include the beautiful imaging and rigorous statistical approaches used by the authors. The ability to assess both axon density and axon function during regeneration is quite powerful. The touch assay adds a unique component to the paper and strengthens the argument that burns are more damaging to sensory structures and that different treatments help to ameliorate this.

A weakness of the study is the lack of genetic and cell-autonomous manipulations. Additional comparisons between transection and burns, in particular with manipulations that specifically modulate ROS generation or cell migration without potentially confounding effects on other cell types or processes would help to strengthen the manuscript.

The use of genetic and cell-autonomous approaches would strengthen our study, however, we were unable to do this due to the lethality of these genetic approaches (or cell autonomous approaches). Basal epithelial migration is necessary for embryonic development. We attempted to circumvent this by generation of larvae transiently expressing a dominant-negative form of Rac, a protein crucial to the migratory process. The chimeric expression of the dominant negative Rac was either damaging to the larvae or the mosaicism was too low to observe any effects on migration phenotype.

We also attempted a genetic approach to manipulate ROS production, as discussed above. We found that the DUOX morpholino was lethal to burned larvae. Finally, we attempted pharmacological inhibition of ROS production using the inhibitor DPI (Diphenyleneiodonium). With this treatment, burned larvae have marginally improved axon density and touch sensitivity, suggesting that dampening ROS may improve outcome. The DPI data have been added to the manuscript.

In terms of framing their results, the authors refer to "sensory neurons" and "sensory axons" throughout the text - it should be made clear what type of neuron(s)/axon(s) are being visualized/assayed. Along these lines, a broader discussion of how burn injuries affect sensory function in other systems - and how the authors' results might inform our understanding of these injury responses - would be beneficial to the reader.

In summary, the authors have established a tractable vertebrate system to investigate different sensory axon wound healing outcomes in vivo that may ultimately allow for the identification of improved treatment strategies for human burn patients. Although the study implicates differences in keratinocyte migration and associated ROS production in sensory axon wound healing outcomes, the links between these processes could be more rigorously established.

The inconsistency between “neuron” and “axon” has been noted and the text has been corrected accordingly. “Neuron” is used when referring to the cell as a whole, while “axon” is used when referring to the sensory processes in the caudal fin. We added information about burn in the introduction as suggested: “While epithelial tissue is well adapted to repair from mechanical damage, burn wounds heal poorly. Thermal injury results in chronic pain and lack of sensation in the affected tissue, suggesting that an abnormal sensory neuron response contributes to burn wound pathophysiology.”

We thank the reviewer’s for their comments.

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

Suggested experiments:

(1) ROS measurements with the dye Pfbsf should be validated with more established ROS probes such as HyPer.

Pfbsf has been used previously as a readout of ROS production, and its use is documented in zebrafish (Maeda et al., *Angew Chem Int Ed Engl*, 2004, and Niethammer et al, *Nature*, 2009). These sources have been added as references when introducing Pfbsf to provide context for its use. The probe was validated and compared to HyPer in Niethammer’s 2009 paper. In our hands, we have used both probes and have similar results with tail transection.

(2) To better support claims on ROS and H2O2 playing a central role in mediating axonal damage, the authors should consider pharmacological approaches such as rescue

experiments with H2O2 and experiments using inhibitors such as DPI or apocynin.

While the above reagents and drugs have limitations and non-specific side effects, more convincing proof could result from genetic approaches including experiments on DUOX knockdown or knockout lines.

To further dissect the role of ROS in the burn response, we conducted experiments using DPI, a potent ROS inhibitor that is well-documented in the literature. We found that 20 μ M treatment of DPI (1 hour pretreatment, 1 hour post-burn) marginally improved axon density when quantified 24 hpw. Any higher dose, when in combination with a burn, proved to be lethal. Longer treatment with DPI was also not tolerated.

In addition to experiments with DPI, we attempted to burn larvae that were injected with DUOX morpholino. The combined use of burn and DUOX MO was lethal. We have dampened the conclusions and include the new data with the DPI in the revised manuscript.

Minor corrections:

(1) A phrase/expression in the abstract is confusing: isotonic treatment does not "induce osmotic regulation". Cells exposed to hypo- or hypertonicity will respond by regulatory volume decrease or increase, respectively. Isotonic treatment maintains homeostasis.

We appreciate this point and agree with the distinction. Revisions have been made in the text accordingly.

(2) Figures 4E and 5E would be better to show as an average of multiple experiments with statistical significance.

The purpose of figures 4E and 5E are to demonstrate changes in fluorescence intensity and localization of ROS using the representative time series shown in 4D and 5D. The figure legend has been updated accordingly.

Reviewer #2 (Recommendations For The Authors):

Figure 3D How can one distinguish between the two cellular elements that randomly meet or that there is actual coordination? Can the interactions be quantified? It is also unclear what the authors mean by "sensory neuron movement". The authors show that the neuronal cell bodies stay in their position, so only the axons change position. Do they do this by growth, i.e. the neuronal growth cones follow the keratinocytes or do keratinocytes displace the axon shafts?

We have included supplemental movies that address this question in the new uploaded document. Figure 3D is comprised of still images taken from supplemental movie 2, which is a timelapse of keratinocytes/axons moving together after a burn injury. This movie clearly shows keratinocytes and their ensheathed axons moving simultaneously, so keratinocytes are mechanically pulling sensory axon shafts with them. We have revised the text to say axon movement, not sensory neuron movement.

Over the time course of axonal movement (1 hour post-burn), it is not possible that neuronal growth cones contribute to movement, as this is too slow – previous work by other labs has shown that it takes several hours for axons to fully regenerate into amputated tissue, with movement not even noticeable until about 3 hours post-wound (Rieger and Sagasti, PLOS Biology, 2011).

Regarding the second point, “neuron” vs. “axon” is an inconsistency in the text that has been corrected. “Neuron” is used when referring to the cell as a whole, “axon” is used when

referring to the processes that innervate the caudal fin. The axons are physically pulled along with keratinocytes as they migrate after burn application. From our observations, growth cones appear closer to the wound site after the movement has stopped.

Figure 4G It is surprising that the visual differences in the distribution of values are not statistically significant.

The distribution of values in 4G was large and that is why there is no statistically-significant difference – we were also surprised at this result. We did all statistics with a statistician and this included rigorous criteria for significance.

Figure 4H The images seem to show a difference, whereas the quantification does not. I suggest choosing more representative images.

Figure 4H has been updated to include a more representative image of axon patterning with CK666 treatment.

Figure 6A The text states that axon damage in the control and isotonic condition is comparable, yet in the image, it appears that the damage in the isotonic treatment at 0 hpw is more distal.

This is a good observation that we consistently see in isotonic-treated fish after burn. Axon damage localizes more proximally in isotonic-treated samples because the keratinocytes distal to the notochord are likely dead, and the axons innervating those cells are likely immediately destroyed upon burn application. As a result, the distal axons are not present to express GCaMP. We believe isotonic treatment allows keratinocytes to live slightly longer, so axon damage is therefore prevented for longer. This is also the focus of continuing work to further understand the burn microenvironment.

Finally, the materials section could mention bias mitigation measures, e.g. withholding the treatment condition from the experimenter in the touch test.

We minimized bias in experiments whenever possible, and the conservative statistical measures that were applied to our data further reduce the likelihood of false significance.

Reviewer #3 (Recommendations For The Authors):

- Line numbers would have facilitated reviewer feedback.
- Supplementary movies were missing in the submission.

The lack of supplementary movies upon submission was a mistake and the movies have been uploaded along with the revised manuscript.

Introduction:

- Pg. 3: "In response to tissue damage, sensory neurons undergo rapid and localized axonal degeneration 4,5." Not sure reference 4 (Reyes et al) is appropriate here as this study was not in the context of tissue damage.

We have revised this section as suggested by the reviewer.

Results:

- The expected expression pattern/localization of several transgenes was unclear. Please clearly state what cell type(s) each should label. For example, pg. 5 - "We next sought to further investigate sensory neuron function in burned tissue. For this, we assessed wound-induced axonal damage using zebrafish larvae that express the calcium probe GCaMP." Where is GCaMP expressed?

The manuscript has been updated to include expression patterns for the included transgenes – in this mentioned case, GCaMP is expressed in neurons under the pan-neuronal *Elavl3* promoter.

- Introducing the GCaMP labeling could use some clarification. Pg. 5 - "As shown previously by other groups, GCaMP labels degenerating neurons in real time³⁵." This is confusing. Do the authors mean that GCaMP increases immediately prior to Wallerian degeneration as shown by Vargas et al. (PMID: 26558774)?

Sustained elevated calcium levels are associated with axon damage. Previous work from other labs has shown that calcium influx follows axon injury (Ziv and Spira, EPN 1993, Adalbert et al., Neuroscience 2012). In these experiments, whenever there are GCaMP-positive punctae, this indicates axon damage. We have revised the manuscript to address this critique.

The *Elavl3*-GCaMP5 transgenic line will label when calcium levels increase in neurons. However, given the parameters used for imaging in our study (20x magnification, 100 ms exposure, and collection speed every 30 seconds for timelapses), we believe that only sufficiently large increases in calcium that are indicative of cell damage, and not physiological function, are being visualized.

- Figure 1E - Are these panels images of the same fish? Please specify in the legend.

Figure 1E is comprised of one transected and one burned larva each, live-imaged over the course of six hours. The legend has been updated to include this information.

- Figure 1F - How was the damage area measured? Consider doing this measurement over time to match Figure 1E.

Axon damage area measurements were performed similar to axon density measurements – maximum intensity z-projected confocal images of the caudal fin were generated using FIJI. For all experiments, the caudal fin area posterior to the notochord was outlined using the Polygon tool and measured to obtain a total surface area ROI. Axon fragments inside the outlined area were manually thresholded so all fragments posterior to the notochord were labeled and no saturated pixels were present, and an area measurement of these thresholded pixels was taken. We have added a section describing these measurements in the Methods section under "Axon damage quantification."

- Pg. 5 - When introducing the *ngn1* MO - please state the expected phenotype and cite the appropriate background literature_.

The *ngn1* morpholino was cited in the Methods section with the appropriate literature (Cornell and Eisen, *Development*, 2002), from which we got the morpholino sequence. We thank the reviewer for pointing out the need for more introduction and clarification in the main text, so the *ngn1* morpholino has been discussed in greater depth and cited in the main text as well using the same citation.

- The two-wound model is an elegant approach but could be more clearly described in the main text.

An improved explanation of the two-wound experiment has been added to the text.

- For Figure 3, it would be helpful to have a schematic of the anatomy illustrating the relative positions of axons and epidermal cell types.

- Figure 3C - should an additional control here be transected? Given that the *krt4:lifeact* transgene labels both layers of the epidermis, how were the superficial and basal keratinocytes separated? Interpretation of this section should be carefully worded. The authors state that "...suggesting that the superficial keratinocytes are being pulled by the motile basal keratinocytes" (pg.7) but isn't another possibility that the superficial cells are stationary?

It is correct that the *krt4:lifeact* transgene labels both layers of keratinocytes, which together span 20-30 microns. These layers were separated from the same z-stack collected by confocal imaging. The first z-slice and last z-slice of the same stack were separated using FIJI and pseudocolored to appear as different colors. This clarification has been added to the Methods.

Prior observations with the *krt4:lifeact* and *krt4:utrch* (figure 3A) transgenic lines reveal that both keratinocyte layers will move distally after burn application.

- Pg. 7 - "The axons of sensory neurons are ensheathed within actin-rich channels running through basal keratinocytes 50,51." ref 51 is a *C. elegans* paper which does not have basal keratinocytes.

This was in error. The correct reference has replaced reference 51 (O'Brien, *J Comp. Neurol.*, 2012), in which electron microscopy is used to document the development of two layers of epithelial cells that also ensheath sensory neurons in a protective manner similar to glial cells in the central nervous system.

- Figures S1E and F - the authors state that RB and DRG soma don't move. However, it was unclear from the figure panels and legend whether the authors imaged neurons that actually innervate the caudal fin (rather than some other region of the animal). Please clarify. For comparison, Fig S1F needs a pre-injury image to be meaningful.

The imaged cell bodies were those in the posterior trunk region, which are responsible for innervating the posterior sections of the fish including the caudal fin. From our observations, there was no movement of neuronal cell bodies after the burn.

- Figure 5 title - can the authors clarify what aspect of this figure relates to "sustained epidermal damage"

The figure 5 title has been updated in response to the reviewer comments.

- Figure 6 - is touch sensitivity really "restored" as the authors suggest? Alternatively, sensitivity may never be lost in isotonic treatment. Or the loss may be delayed?

We have modified the text accordingly by updating our phrasing – "restored" has been replaced with "improved" to indicate benefit over time.

- Can the authors further disentangle the effects of keratinocyte migration, ROS, and isotonic treatment on axon regeneration? For example, would the addition of CK666 to

the Isotonic +1 hpw treatment improve axon regeneration? Can the authors directly manipulate ROS signaling (e.g., through exogenous addition of H2O2 or duox1 MO) to alter regeneration outcomes in their wounding assays?

See the comments above.

- Figure 6 title - consider removing or clarifying the word "excessive" here

The title has been revised according to the reviewer suggestion.

- hpw vs hpb were used inconsistently throughout the text

The manuscript has been revised to use “hpw” when referring to the timeframe after injury application.

Methods:

- Zebrafish transgenics are missing allele names

References:

- Many mistakes were noted in this section e.g., journal names missing, wrong authors, typos, DOIs misformatted

The references section has been corrected to use formatting consistent with APA citation and eLife preferred guidelines.

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